



# From Qubits to Quantum AI: Introduction to Quantum Computing

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September 17 2026

# Quantum Computing - Beginnings

Quantum Computers – first proposed by Prof. Richard Feynman in 1981 as a new way of computing inspired by nature



## Simulating Physics with Computers

Richard P. Feynman

*Department of Physics, California Institute of Technology, Pasadena, California 91107*

*Received May 7, 1981*

### 1. INTRODUCTION

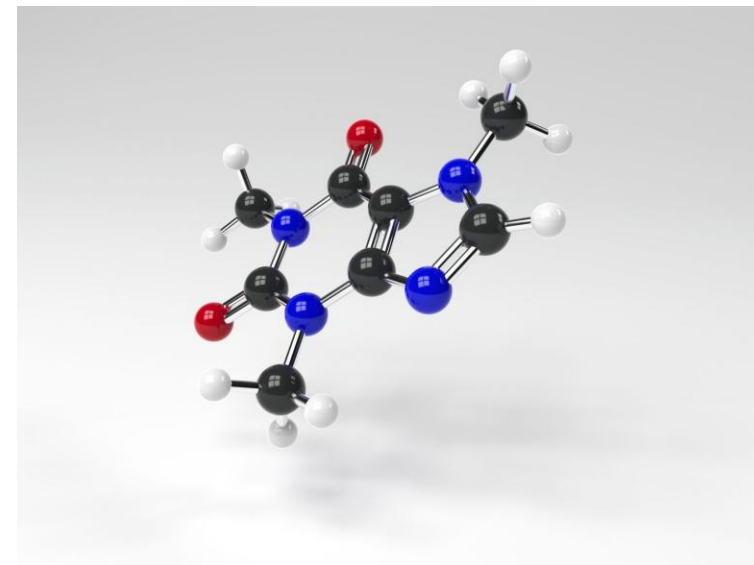
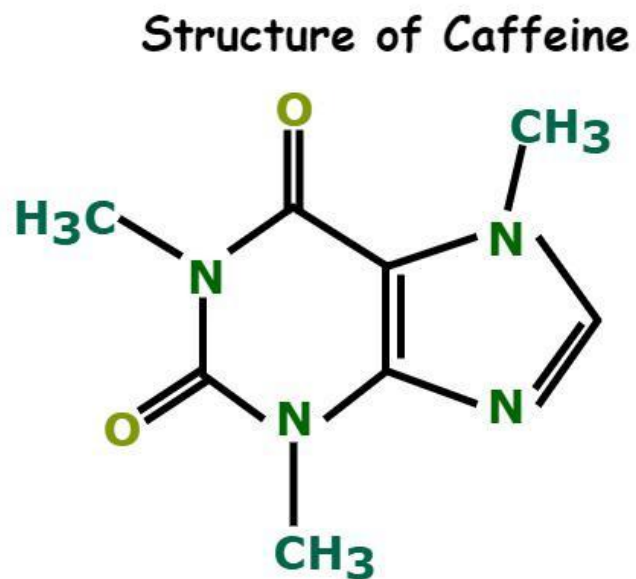
On the program it says this is a keynote speech—and I don't know what a keynote speech is. I do not intend in any way to suggest what should be in this meeting as a keynote of the subjects or anything like that. I have my own things to say and to talk about and there's no implication that anybody needs to talk about the same thing or anything like it. So what I want to talk about is what Mike Dertouzos suggested that nobody would talk about. I want to talk about the problem of simulating physics with computers and I mean that in a specific way which I am going to explain. The reason for doing this is something that I learned about from Ed Fredkin, and my entire interest in the subject has been inspired by him. It

### 4. QUANTUM COMPUTERS—UNIVERSAL QUANTUM SIMULATORS

The first branch, one you might call a side-remark, is, Can you do it with a new kind of computer—a quantum computer? (I'll come back to the other branch in a moment.) Now it turns out, as far as I can tell, that you can simulate this with a quantum system, with quantum computer elements. It's not a Turing machine, but a machine of a different kind. If we disregard the continuity of space and make it discrete, and so on, as an approximation (the same way as we allowed ourselves in the classical case), it does seem to

# Quantum Computing - Beginnings

Nature routinely computes on an extremely large scale using quantum mechanical principles – why can't we?



Number of possible configurations of caffeine molecule:  $2^{242} \sim 10^{73}$

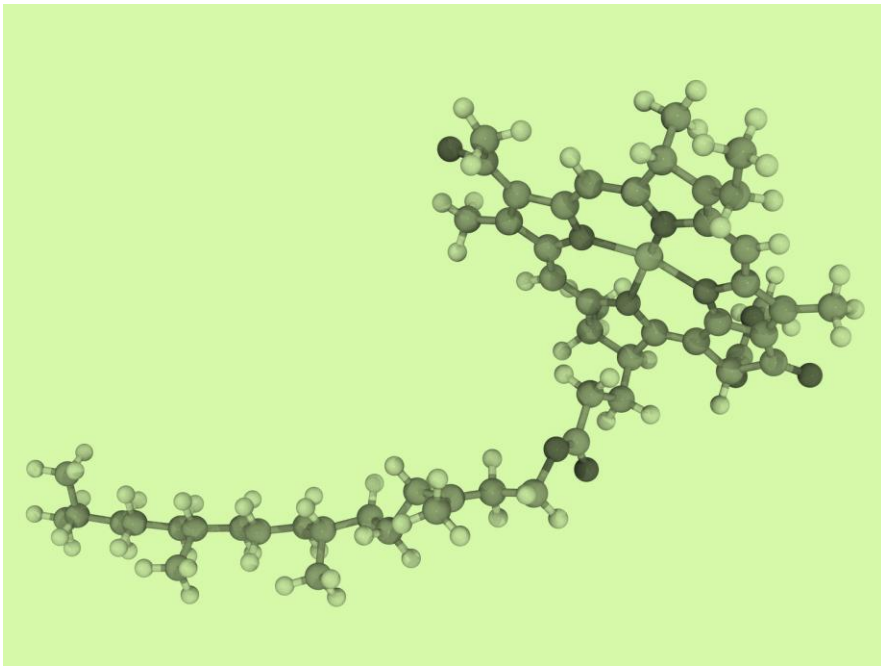
Number of bits on conventional computers to simulate caffeine:  $10^{48}$

Number of stars in the universe:  $\sim 10^{24}$

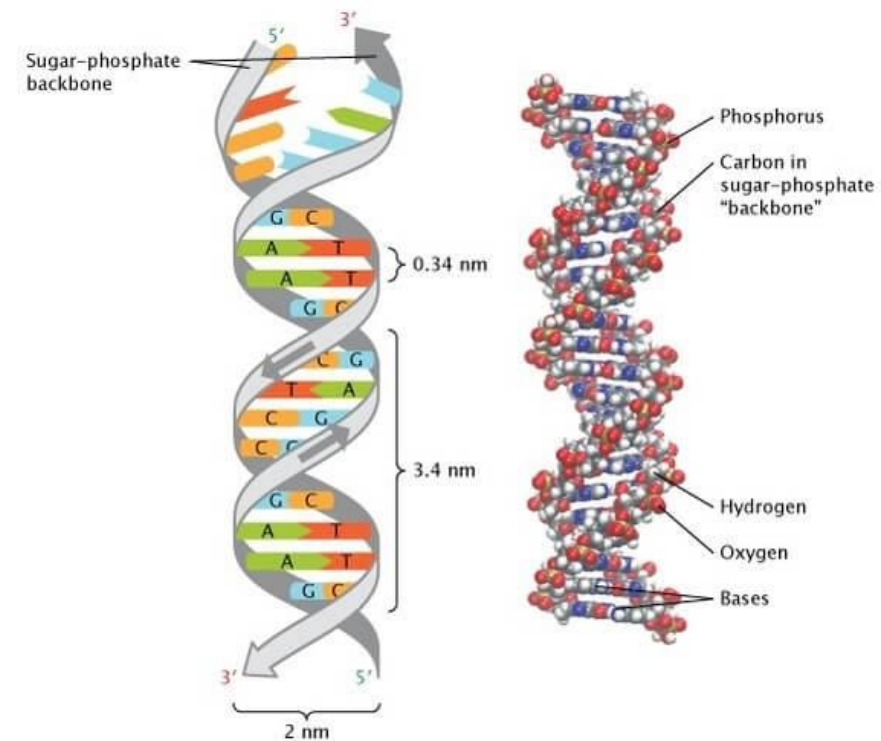
Number of grains of sand on all the beaches on Earth:  $7.5 \times 10^{18} - 21$

# Quantum Computing - Beginnings

Nature routinely computes on a much large scale using quantum mechanical principles – why can't we?



**Chlorophyll molecule**



**DNA**

# Quantum Computing - Beginnings

Key quantum principles – can we use them in computation?

## 1. Everything is Energy

At the deepest level, all matter is made of vibrating energy. Solid objects are not truly solid — they're fields of movement.

## 2. Wave-Particle Duality

Particles behave as both waves and particles. Reality exists as possibility until it's observed.

## 3. The Observer Effect

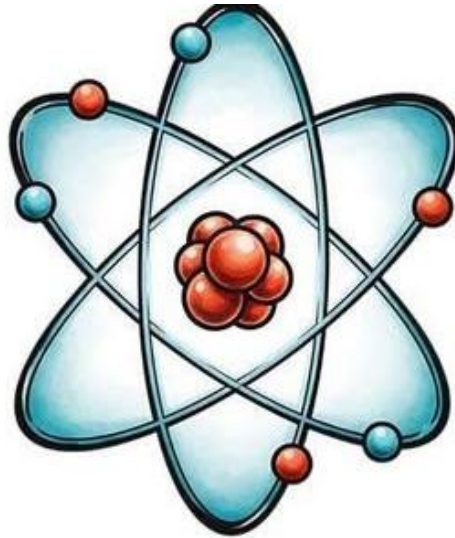
The act of observation changes the outcome. Conscious attention influences physical reality.

## 4. Superposition

A particle can exist in multiple states at once. Reality holds many potential versions simultaneously.

## 5. Quantum Entanglement

Particles can remain connected across vast distances. A change in one instantly affects the other.

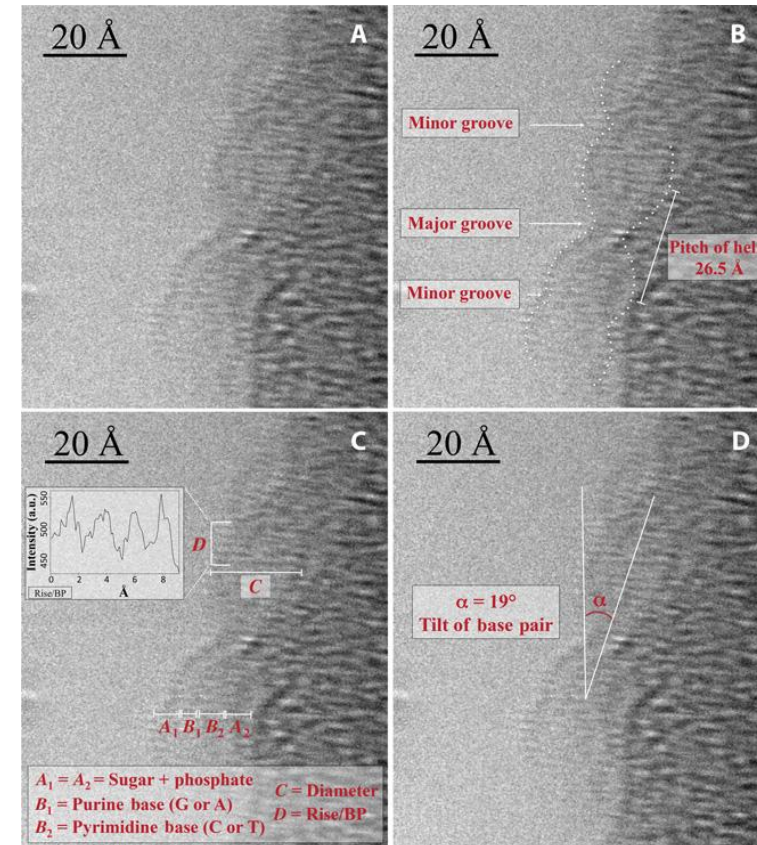


## 6. Uncertainty Principle

You can't know everything about a system at the same time. The universe has built-in limits to certainty and control.

## 7. Probability, Not Certainty

Events don't happen deterministically — only probabilistically. Reality is a field of likelihoods, not fixed outcomes.



Scanning Electron Microscope (SEM) of DNA

# Quantum Computing – Deutsch & Jozsa

Prof. David Elieser Deutsch and Prof. Richard Jozsa formulated the first concrete principles underlying modern quantum computers and created the very first quantum algorithm



## Quantum theory, the Church-Turing principle and the universal quantum computer

DAVID DEUTSCH\*

Appeared in *Proceedings of the Royal Society of London A* **400**, pp. 97-117 (1985)<sup>†</sup>

(Communicated by R. Penrose, F.R.S. — Received 13 July 1984)

### Abstract

It is argued that underlying the Church-Turing hypothesis there is an implicit physical assertion. Here, this assertion is presented explicitly as a physical principle: 'every finitely realizable physical system can be perfectly simulated by a universal model computing machine operating by finite means'. Classical physics and the universal Turing machine, because the former is continuous and the latter discrete, do not obey the principle, at least in the strong form above. A class of model computing machines that is the quantum generalization of the class of Turing machines is described, and it is shown that quantum theory and the 'universal quantum computer' are compatible with the principle. Computing machines resembling the universal quantum computer could, in principle, be built and would have many remarkable properties not reproducible by any Turing machine. These do not include the computation of non-recursive functions, but they do include 'quantum parallelism', a method by which certain probabilistic tasks can be performed faster by a universal quantum computer than by any classical restriction of it. The intuitive explanation of these properties places an intolerable strain on all interpretations of quantum theory other than Everett's. Some of the numerous connections between the quantum theory of computation and the rest of physics are explored. Quantum complexity theory allows a physically more reasonable definition of the 'complexity' or 'knowledge' in a physical system than does classical complexity theory.

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<sup>†</sup>This version (Summer 1991) was edited and converted to L<sup>A</sup>T<sub>E</sub>X by Wim van Dam at the Centre for Quantum Computation. Email: wim.vandam@qubit.ox.ac.uk



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## Rapid Solution of Problems by Quantum Computation

David Deutsch and Richard Jozsa

*Proc. R. Soc. Lond. A* 1992 **439**, 553-558  
doi: 10.1098/rspa.1992.0167

### References

**Article cited in:**  
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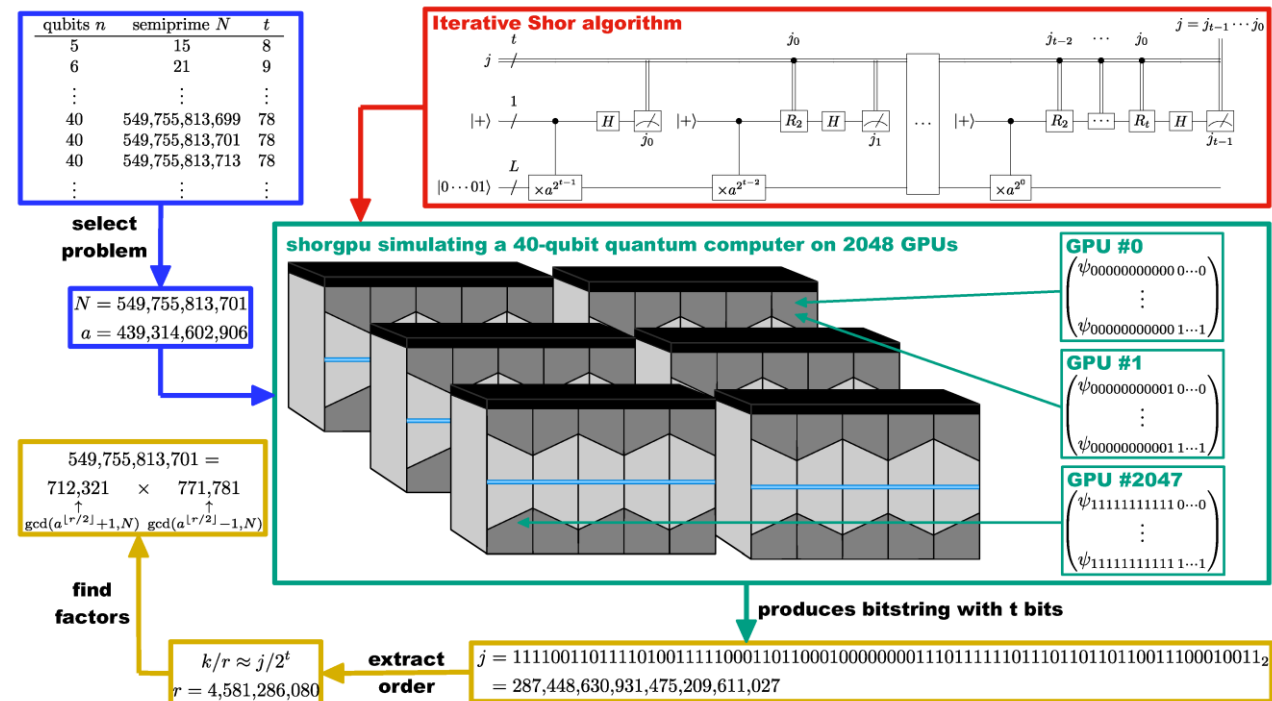
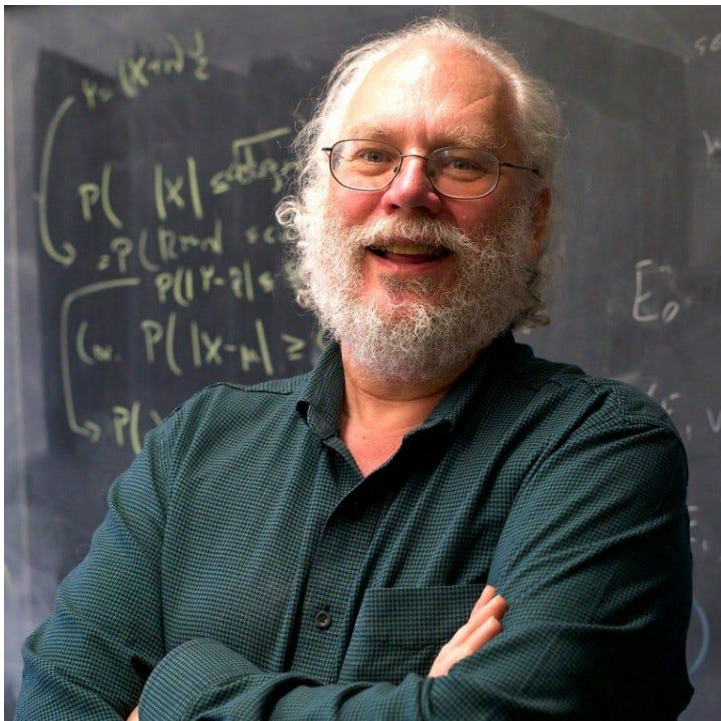
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Quantum theory of computation and construction of the universal quantum computer (1984)

Deutsch-Jozsa Quantum Algorithm (1992)

# Quantum Computing – Shor's Algorithm

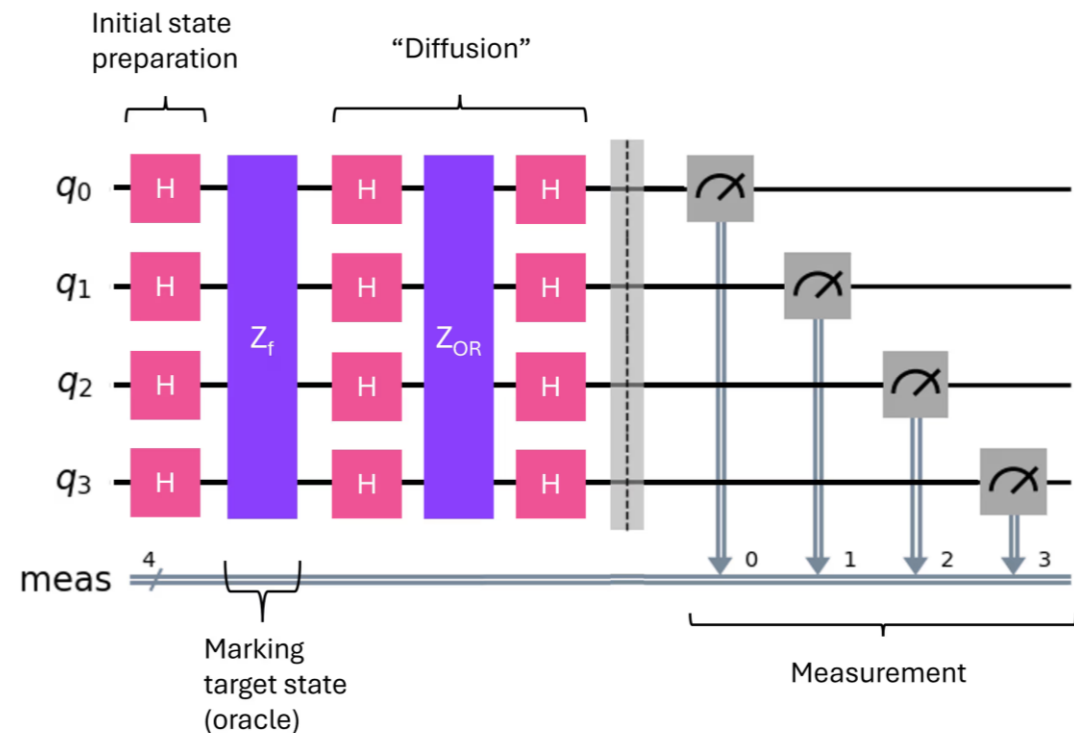
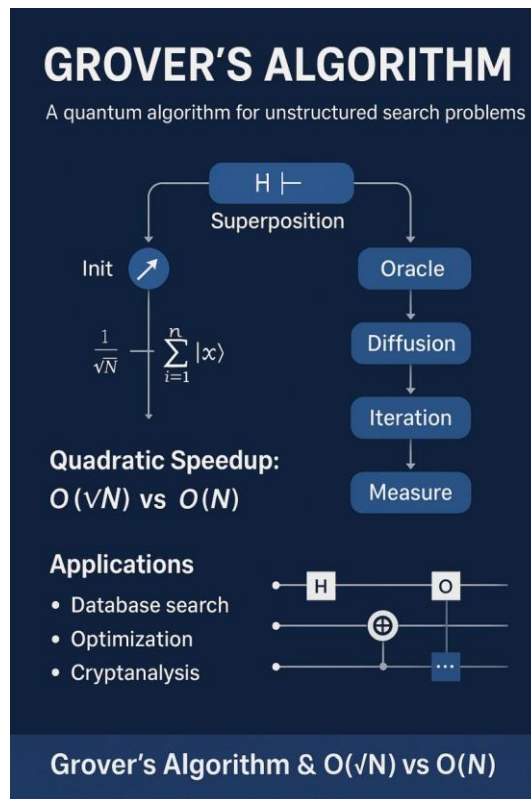
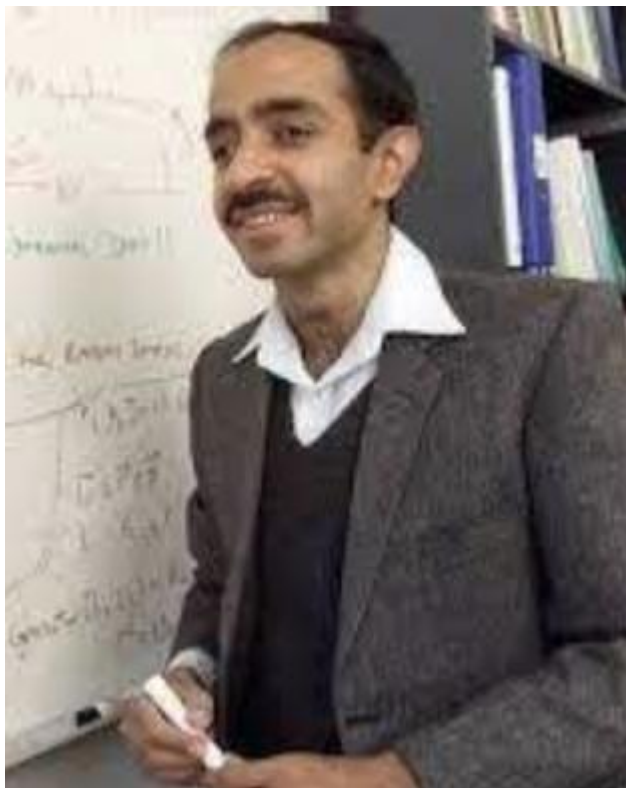
Prof. Peter Shor invented “Shor's Algorithm” a very fast quantum computing algorithm in 1994 to factorize any number into primes with potential to break modern encryption methods (RSA -128/256 bit encryption)



Largest number factorized 39-bit number: 549,755,813,701  
 (Simulating Shor's algorithm running on a quantum computer on a supercomputer)

# Quantum Computing – Grover's Algorithm

Dr. Lov Kumar Grover (IIT Delhi B.Tech. 1981) proposed Grover's Algorithm, which can search a database faster than a conventional search (1996), also useful in breaking encryption methods



**Grover's Algorithm**

# Quantum Computing: First Quantum Computers

1998: First experimental quantum computer: 2-qubit nuclear magnetic resonance (NMR) quantum computer running Deutsch-Josza algorithm

2011: First commercial quantum computer: D-Wave One (sold to Lockheed Martin)

Implementation of a Quantum Search  
Algorithm on a Nuclear Magnetic Resonance  
Quantum Computer

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<sup>\*</sup> Oxford Centre for Molecular Sciences, New Chemistry Laboratory,  
South Parks Road, Oxford, OX1 3QT, UK,

<sup>†</sup> Centre for Quantum Computation, Clarendon Laboratory, Parks  
Road, Oxford OX1 3PU, UK

<sup>‡</sup> Mathematical Institute, 24-29 St Giles', Oxford, OX1 3LB, UK

The simulation of quantum mechanical systems with classical computers appears to be a computationally intractable problem. In 1982 Feynman<sup>1</sup> reversed this observation, suggesting that quantum mechanical systems have an information processing capability much greater than that of corresponding classical systems, and thus could be used to implement a new type of powerful computer. In 1985 Deutsch<sup>2</sup> described a quantum mechanical Turing machine, showing that quantum computers could indeed be constructed. Since then there has been extensive research in this field, but while the theory is fairly well understood actually building a quantum computer has proved extremely difficult, and only two methods have been used to demonstrate quantum logic gates: ion traps,<sup>3,4</sup> and nuclear magnetic resonance (NMR).<sup>5,6</sup> NMR quantum computers have previously been used to demonstrate quantum algorithms to solve the two bit Deutsch problem.<sup>7,8</sup> Here we show how such a computer can be used to implement a fast quantum search algorithm initially developed by Grover.<sup>9,10</sup>

Among other applications Grover's algorithms enable an extremely rapid search over the domain of a binary function to find elements for which this function is satisfied (that is, the function has the value 1). This approach is simpler if the number of satisfying values is known beforehand, and is particularly simple when precisely one quarter of the elements in the domain satisfy the function.<sup>11</sup> The algorithm can be demonstrated using a computer with two quantum bits (qubits) to search a two bit domain in which one of the four elements satisfies the function. A classical search of this domain would require between 1 and 3



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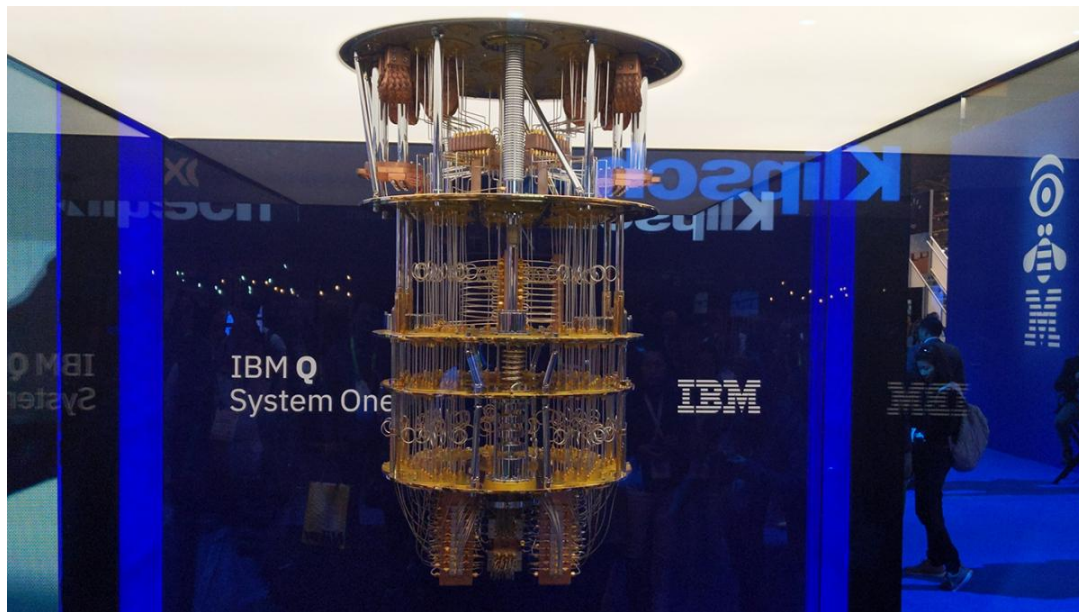


D-Wave One™  
information

# IBM System One - 2019

In 2019, IBM released IBM System One, the world's first integrated, circuit-based commercial quantum computer designed for use outside of research labs

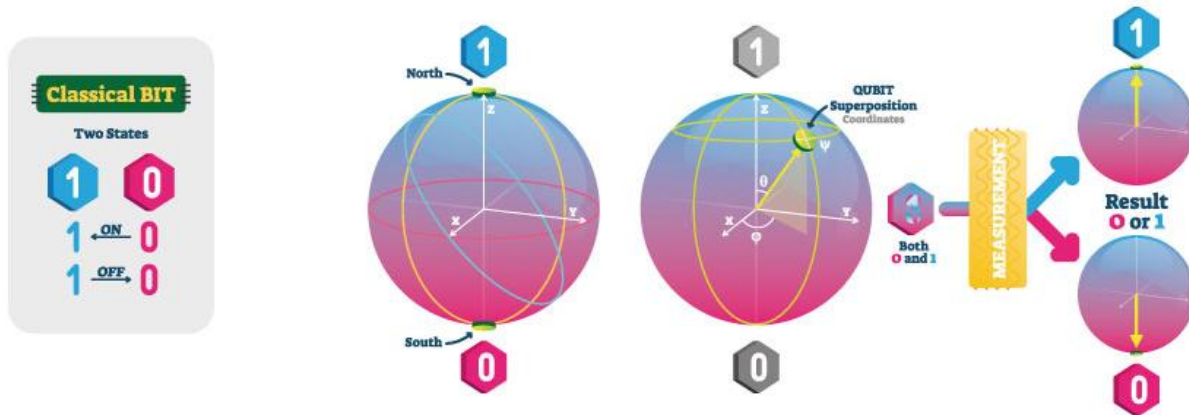
In parallel IBM released “Qiskit” (*Quantum Information Science Kit*), a Python based SDK to develop applications  
Made available IBM System Ones to any developers on the IBM Quantum Platform





# Quantum Computing Fundamentals (Some Mathematics Required)

# Qubits

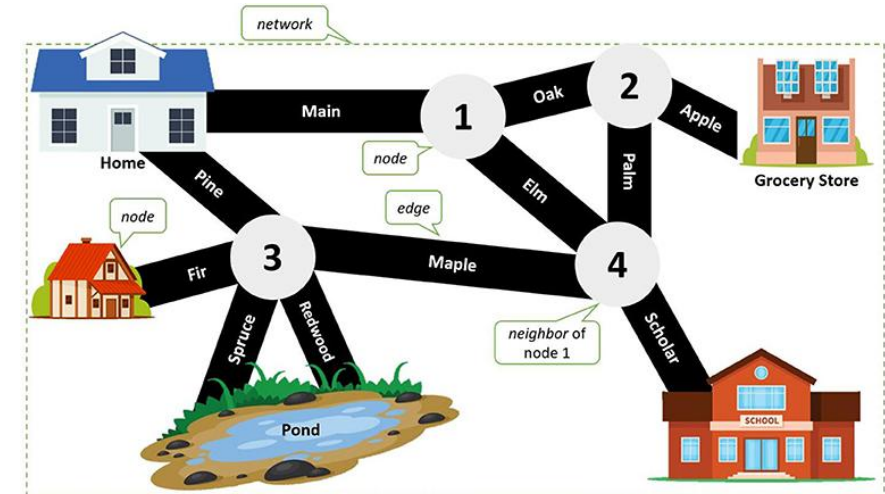


## Quantum Superposition

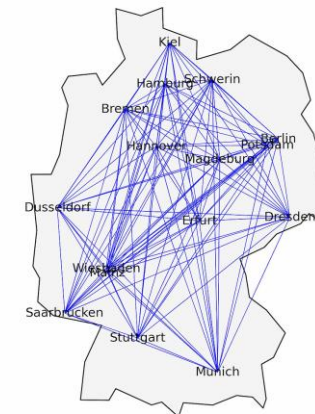
Qubits exist in multiple states → can store more information

## Quantum Entanglement

Qubits can be interconnected even if physically distant → information grows exponentially when scaling in a quantum system:  **$N$  bits in classical can be represented by  $2^N$  qubits in quantum**

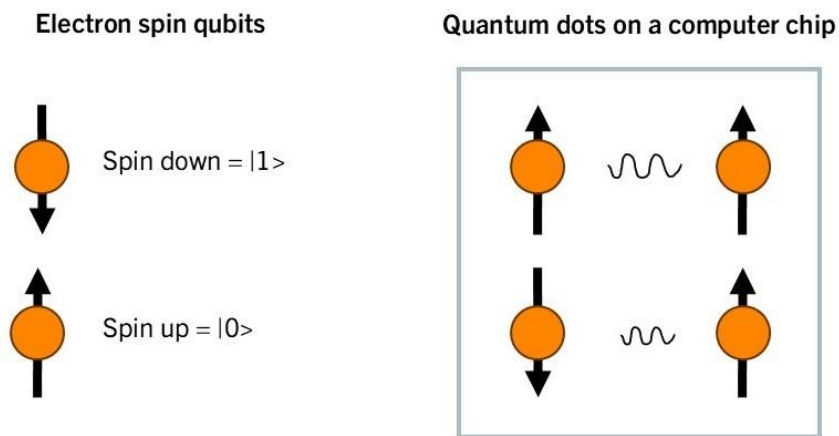


All possible paths between any two state capitals in Germany



# Bra, Ket and Tensors

## “Bra” and “Ket” Representation of a qubit



$$\text{Bra: } \langle 0| = [1 \ 0], \langle 1| = [0 \ 1]$$

$$\text{Ket: } |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\text{Unknown spin: } |\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

## Combining qubits using tensor operations

$$|10\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

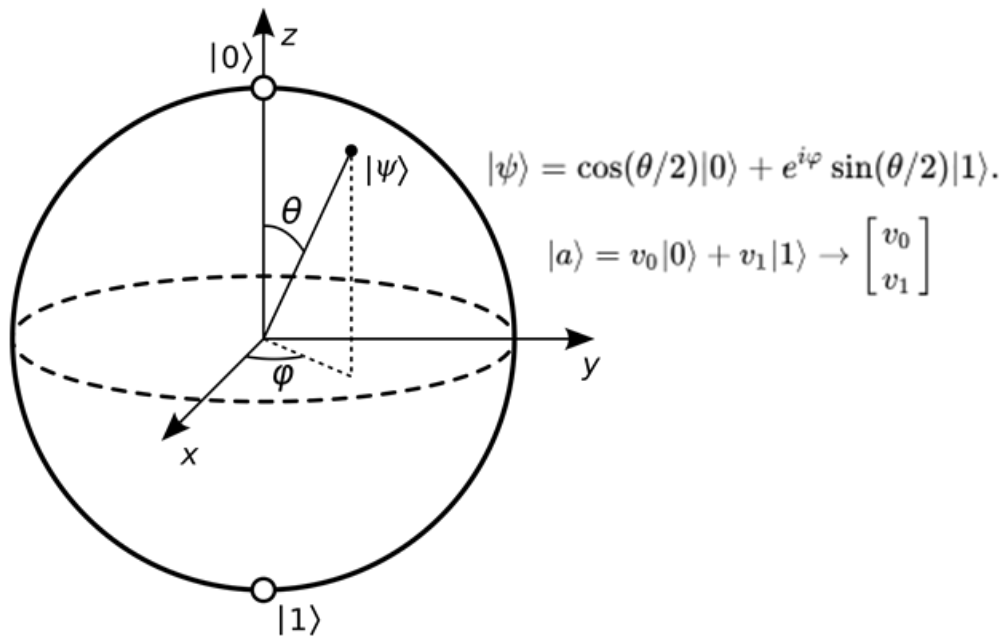
$$|01\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|11\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

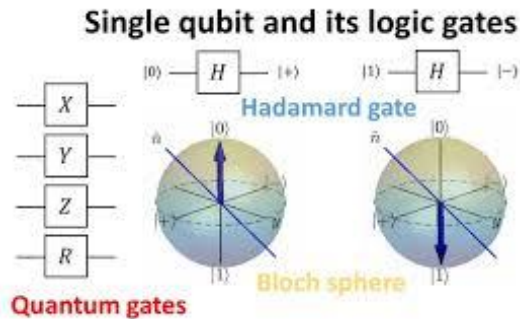
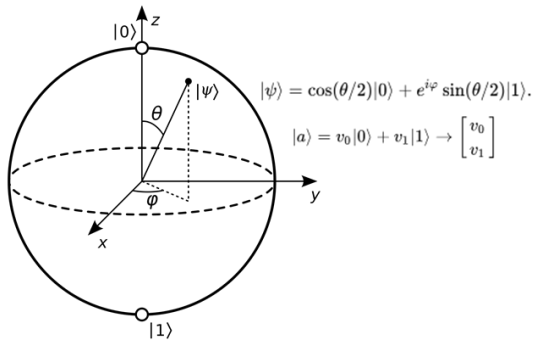
# Quantum Gates

Quantum Gates operate on one or more Qubits to change their state  
 Qubits represented on a “Bloch sphere”, or as vectors (“kets”)  
 Represented by matrix operations, vector and tensor products



| Operator                   | Gate(s)                                                                                                                                                                 | Matrix                                                                                                                                                                                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pauli-X (X)                |   | $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$                                                                                                                                                                                                                                                       |
| Pauli-Y (Y)                |                                                                                      | $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$                                                                                                                                                                                                                                                      |
| Pauli-Z (Z)                |                                                                                      | $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$                                                                                                                                                                                                                                                      |
| Hadamard (H)               |                                                                                      | $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$                                                                                                                                                                                                                                   |
| Phase (S, P)               |                                                                                      | $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$                                                                                                                                                                                                                                                       |
| $\pi/8$ (T)                |                                                                                      | $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$                                                                                                                                                                                                                                              |
| Controlled Not (CNOT, CX)  |                                                                                      | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$                                                                                                                                                                                                     |
| Controlled Z (CZ)          |                                                                                    | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$                                                                                                                                                                                                    |
| SWAP                       |                                                                                    | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$                                                                                                                                                                                                     |
| Toffoli (CCNOT, CCX, TOFF) |                                                                                    | $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ |

# Single and Two Qubit Gates



$$\text{Pauli}_x: X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_2 \\ v_1 \end{bmatrix}$$

$$\text{Pauli}_y: Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -iv_2 \\ +iv_1 \end{bmatrix}$$

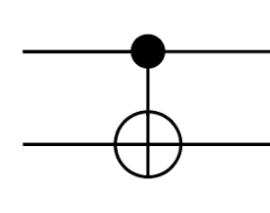
$$\text{Pauli}_z: Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ -v_2 \end{bmatrix}$$

Hadamard:  $H|a\rangle$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}}(v_1 + v_2) \\ \frac{1}{\sqrt{2}}(v_1 - v_2) \end{bmatrix}$$

Controlled NOT or *CNOT* gate

The second bit flips if the first one is 1



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\text{CNOT } |10\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle$$

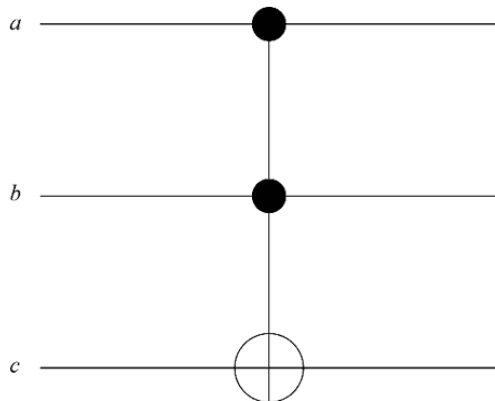
$$\text{CNOT } |01\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = |01\rangle$$

$$\text{CNOT } |11\rangle = |10\rangle$$

$$\text{CNOT } |00\rangle = |00\rangle$$

# Three Qubit Gates

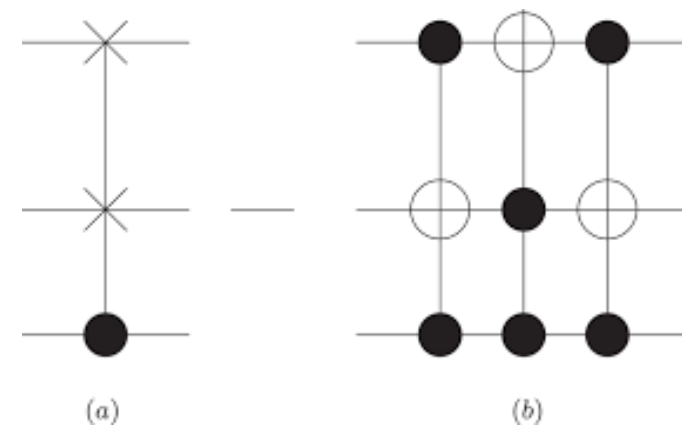
Toffoli Gate



The first two qubits are the control qubits and remain unchanged, If both of them are identical, then the second qubit is modified with a NOT operation

$T |110\rangle = |111\rangle$  : both control qubits are  $|1\rangle$  hence third qubit flips,  $T |101\rangle = |101\rangle$  : control qubits are  $|1\rangle$  and  $|0\rangle$ , hence the third qubit remains the same

Friedkin Gate

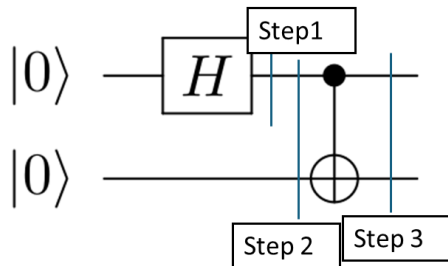


The first qubit is the control qubit  
If the first qubit is  $|1\rangle$  then the second and third qubits are swapped

$F |110\rangle = |101\rangle$  : control qubit is  $|1\rangle$  hence second and third qubits swap  
 $F |001\rangle = |101\rangle$  : control qubits is  $|0\rangle$ , hence they don't swap

# Quantum Entanglement

Quantum Entanglement: Two qubits are connected together, and a change in one changes the other, irrespective of distance (“spooky action at a distance”)



Step 1: Apply Hadamard Gate to first (primary) qubit

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \alpha + \beta \\ \alpha - \beta \end{bmatrix}$$

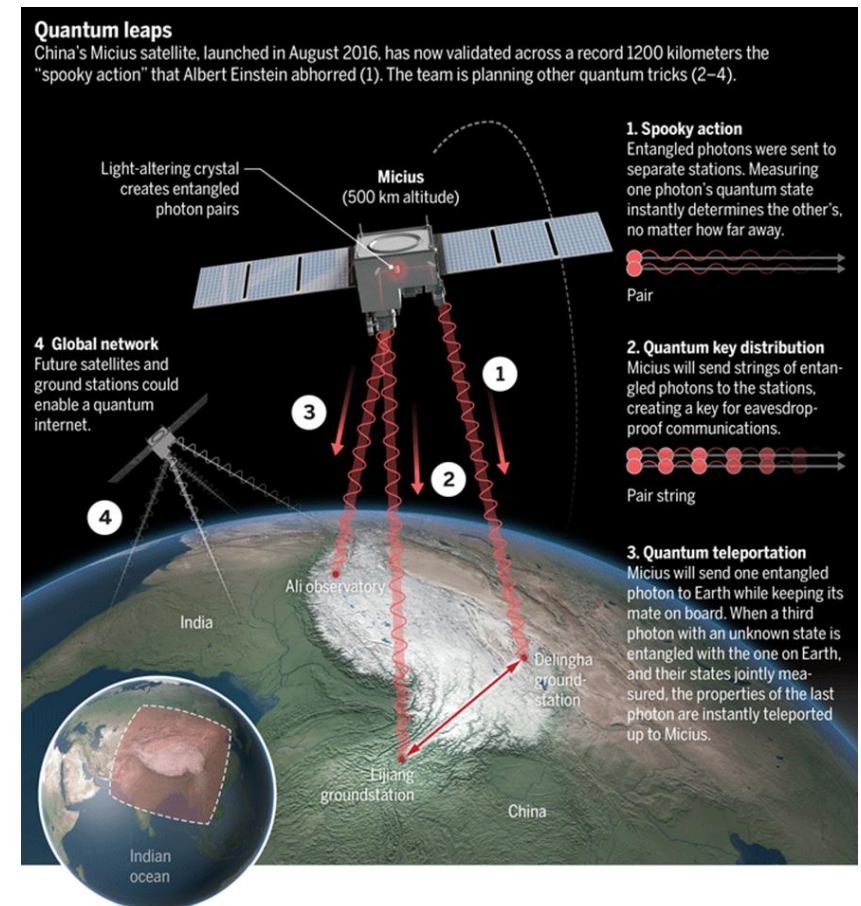
$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 2: Take Tensor Product with second (control) qubit

$$|H|0\rangle|0\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |10\rangle$$

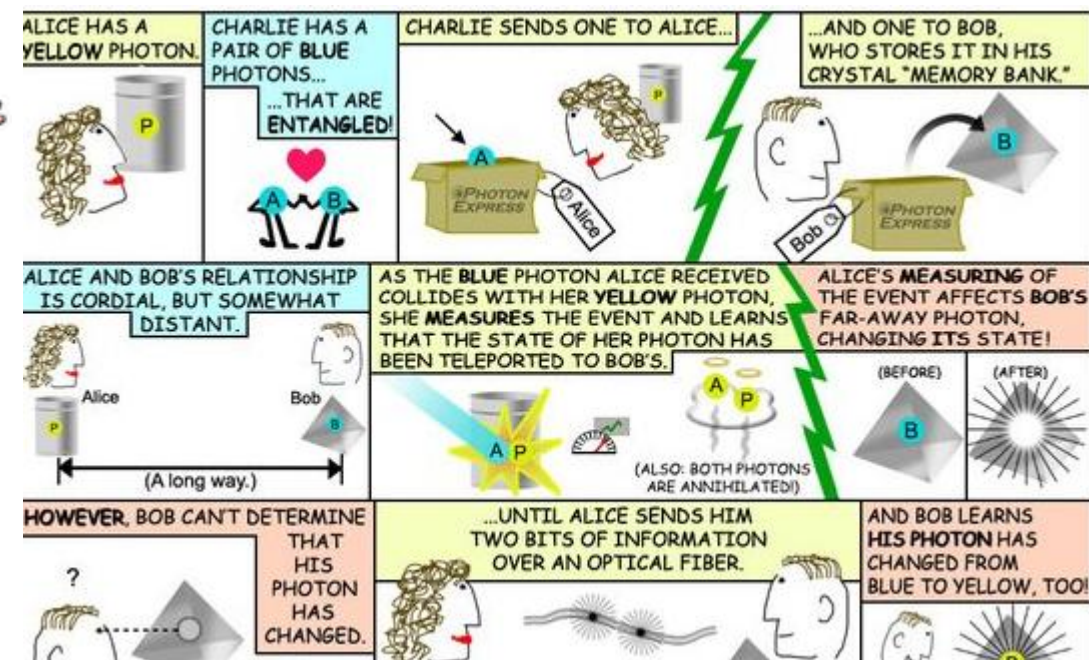
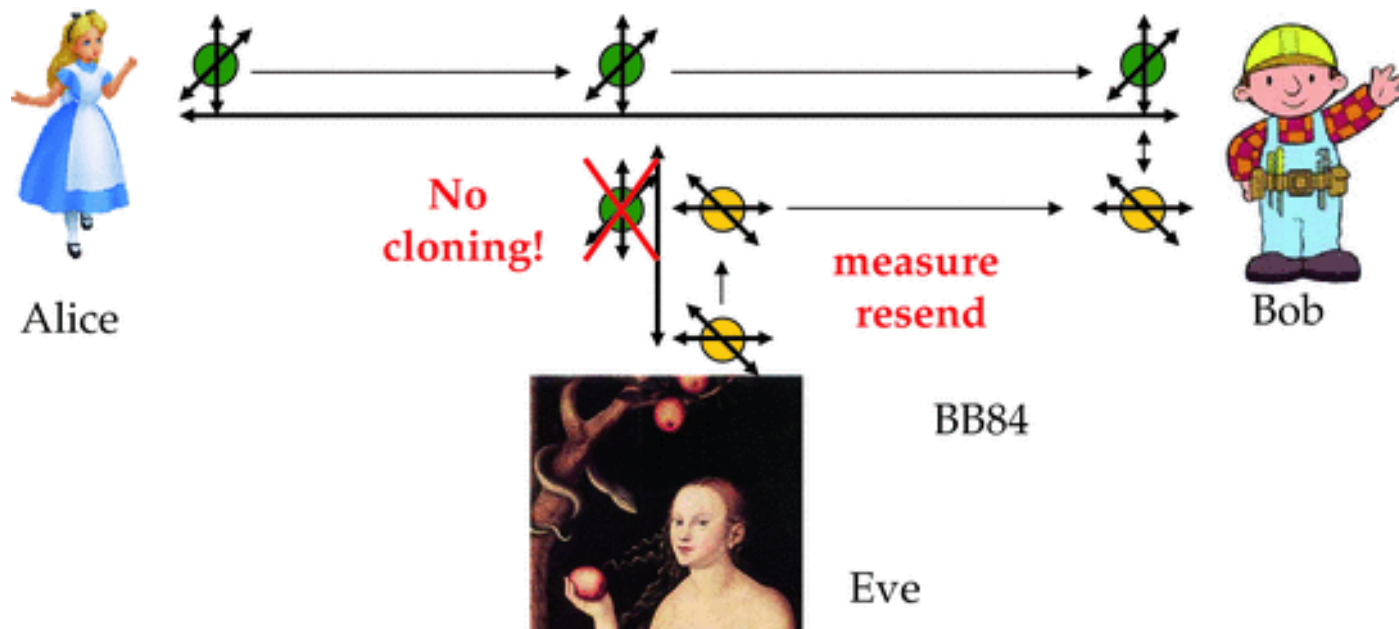
Step 3: Apply CNOT gate to tensor product

$$\text{CNOT} |H|0\rangle|0\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$



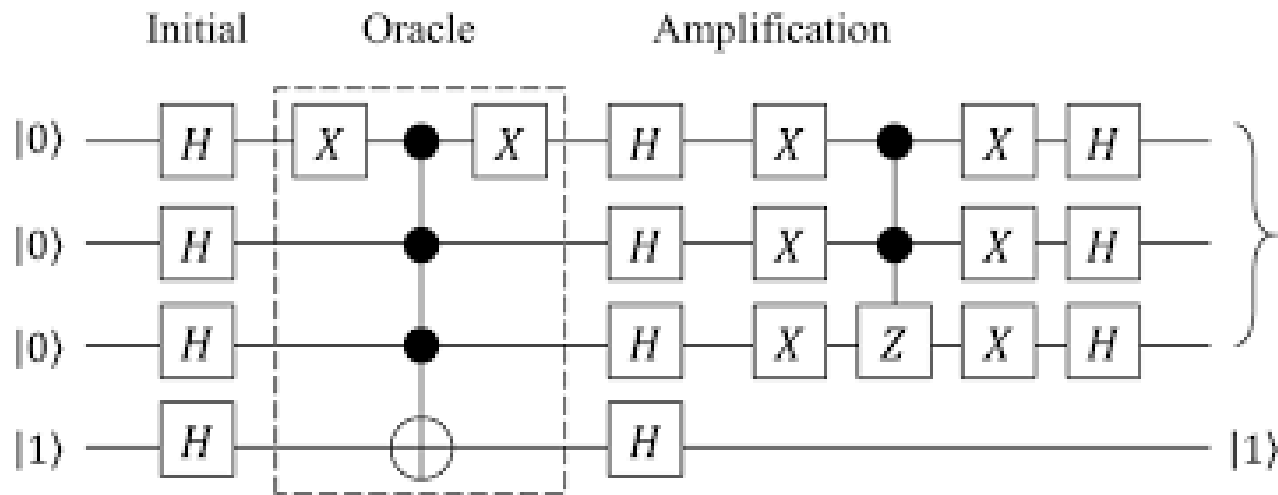
# No Cloning

Qubits cannot be intercepted or cloned - creates a new tool for secure communication  
Quantum teleportation possible!

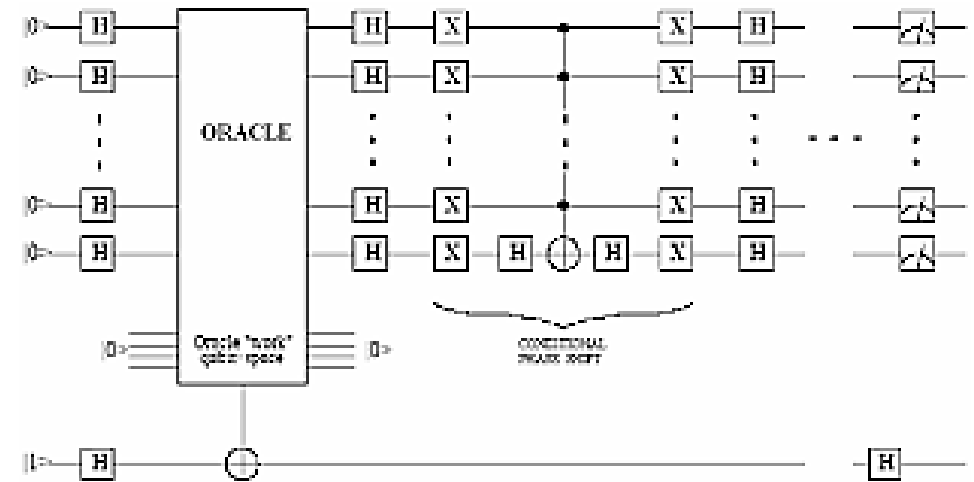


# Quantum Circuits

- Sequence of gates, entanglements on multiple qubits, which encode any kind of information and measure at the end
- Designed to accomplish a specific task, hence they are also algorithms
- A new toolkit for computation and information processing
- Can be faster than or accelerate conventional algorithms for the same task



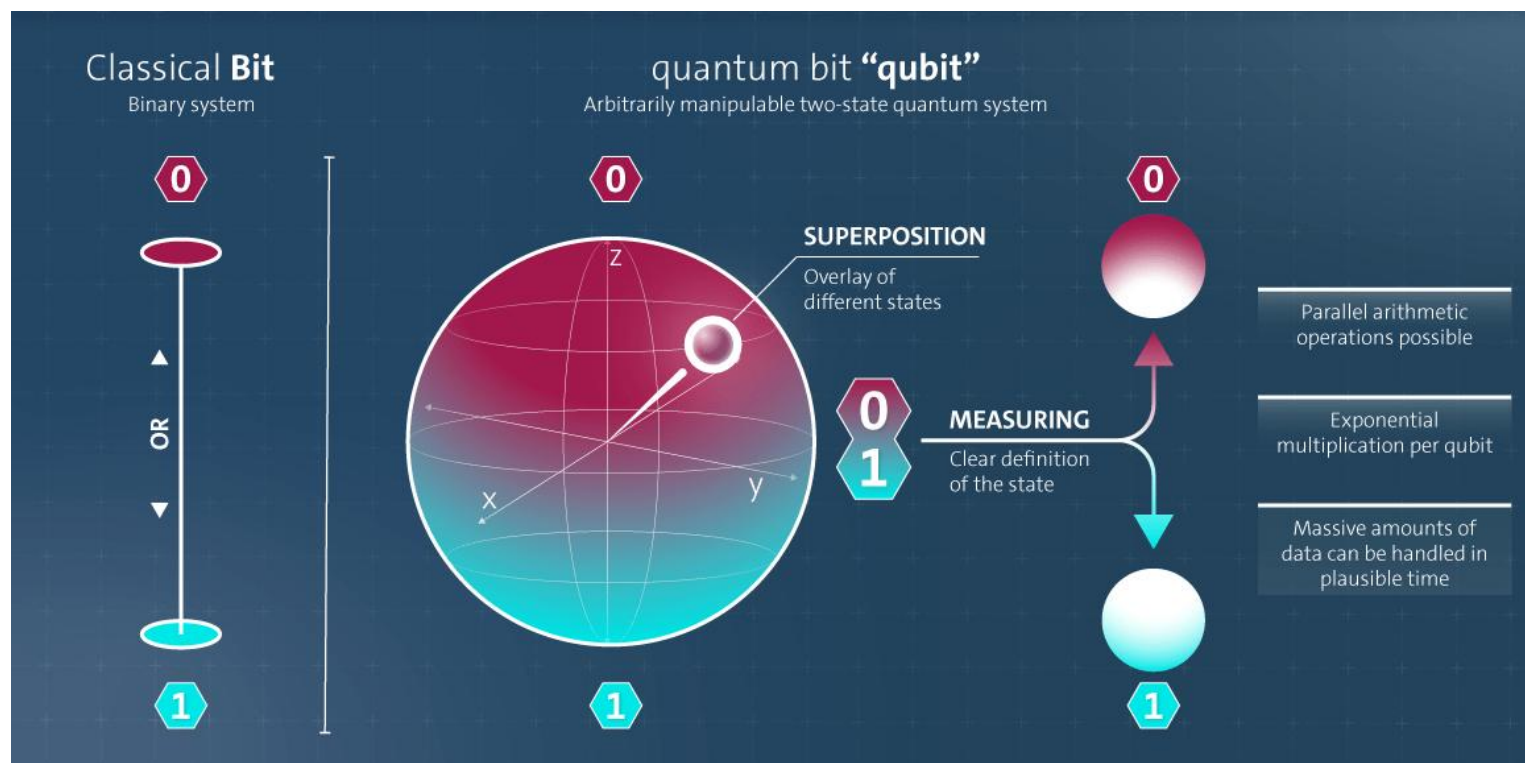
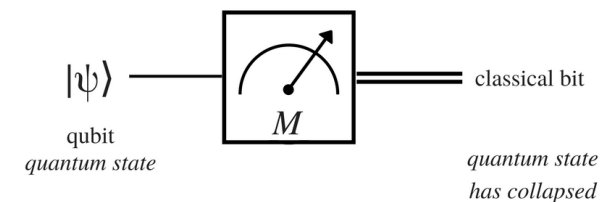
Grover's Search Algorithm: Circuit for four qubits  
Fast search algorithm ( $O(\sqrt{N})$ ) vs



Grover's Search Algorithm for Any number of qubits  
Potential to break 128-bit/256-bit encryption

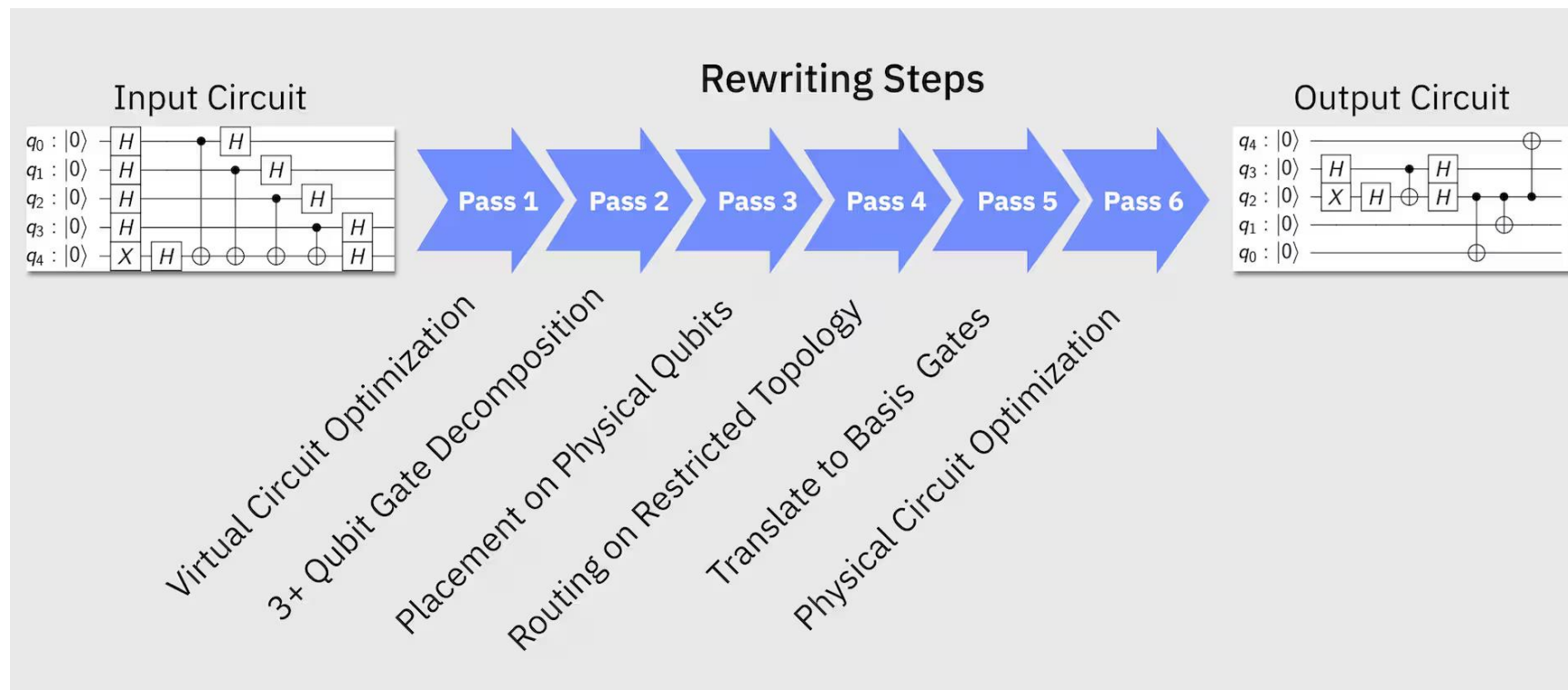
# Qubit Measurements

- Qubits are in superposition taking all possible values
- The measurement step makes qubits collapse to a final definite value
- Typically final step in a quantum circuit



# Transpilation

Transpilation: conversion of quantum circuits to hardware level circuit execution on physical qubits and gates  
Gate depth: Optimize both given quantum circuit and the physical output circuit to reduce the number of gates  
Shots: On the hardware, the physical circuit is run multiple times to average out probabilistic behavior





DEMO

# Quantum System Development Tools

Many frameworks and libraries available to develop quantum algorithms

Describe quantum algorithms at high level in the framework using libraries

Simulate the operations using emulators to ensure correct results

Transpile to quantum operations for specified target hardware (e.g. Qiskit – IBM, Cirq – Google)

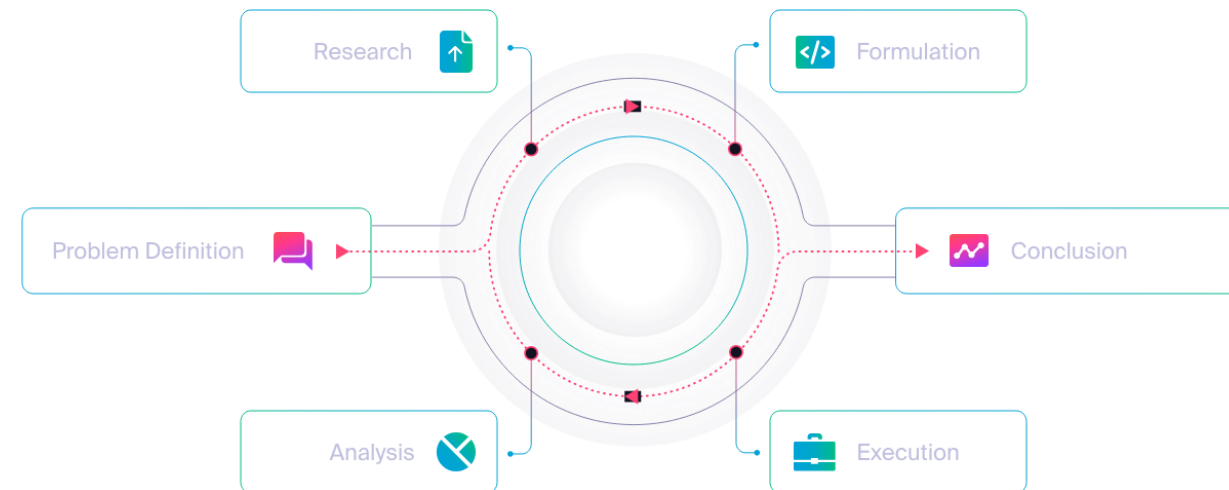
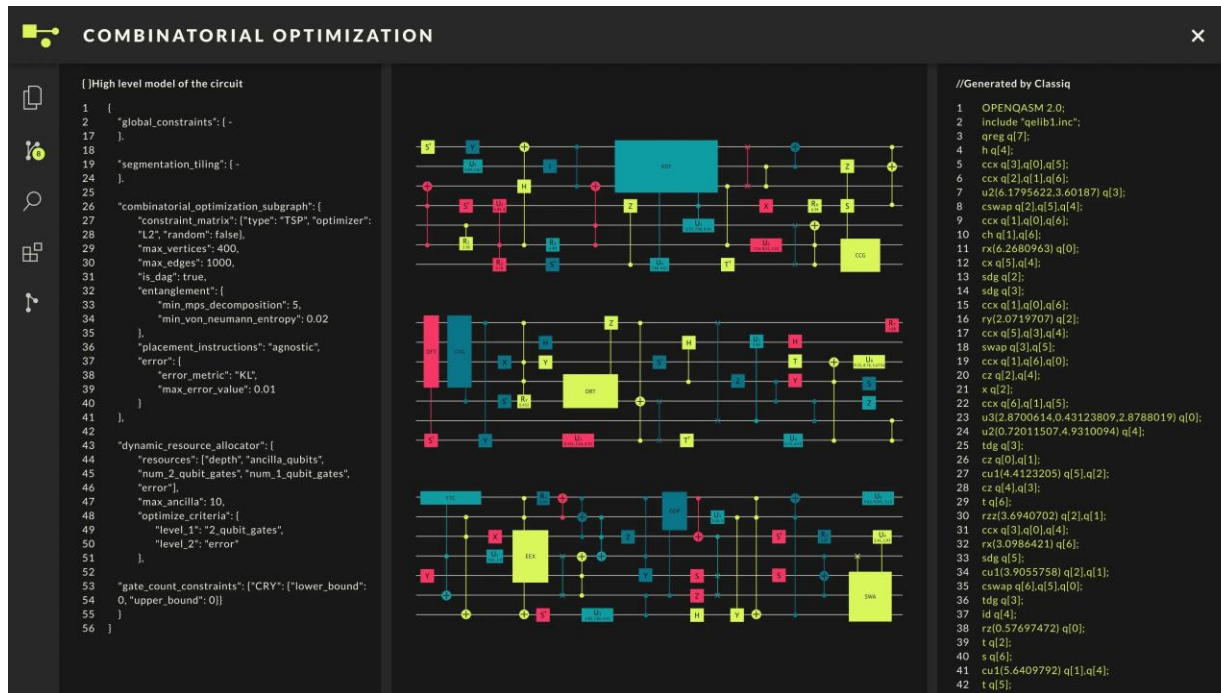


# Classiq and Strangeworks

General purpose environments and toolsets to democratize quantum computing for all QPUs and applications

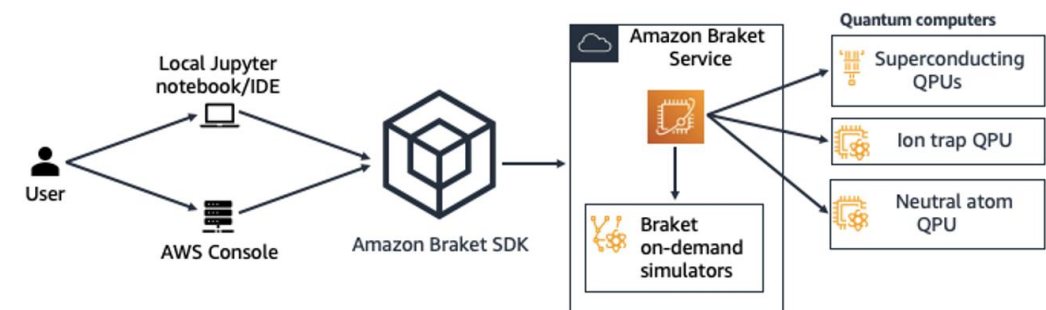
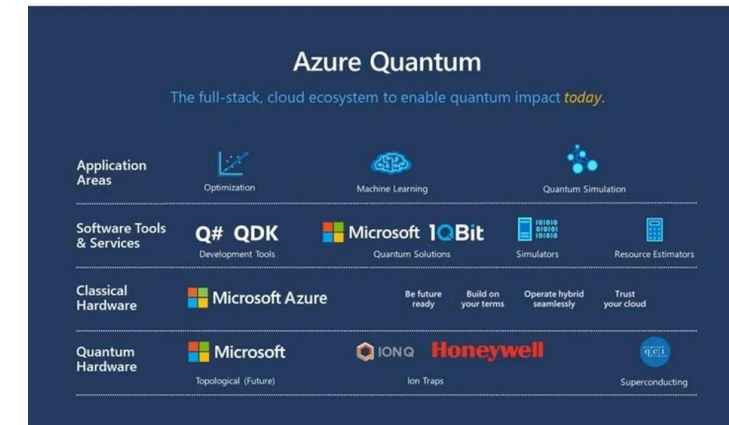
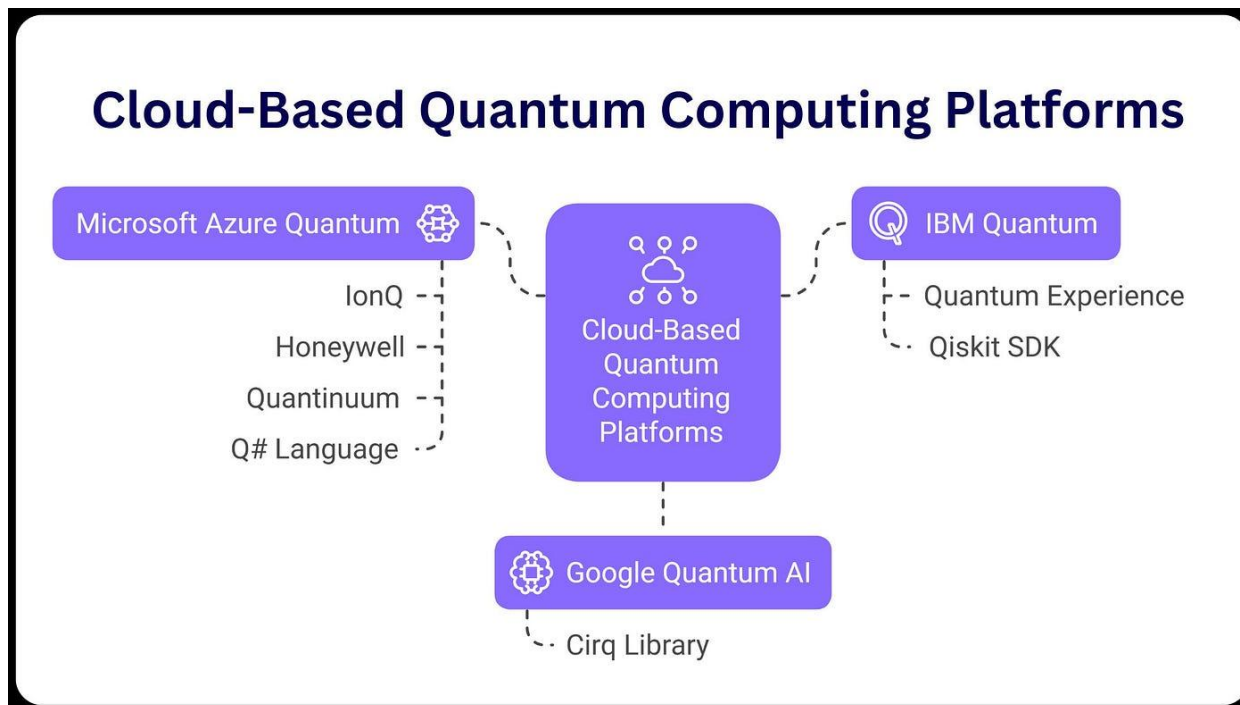
Classiq: IDE environment with SDK (Software Development Kit) for quantum algorithm development

Strangeworks: Cloud based hardware agnostic platform to develop and deploy quantum solutions using workflows



# Quantum Computing Cloud Platforms

- Access quantum computers without installing and paying for a quantum computer
- Accompanied by SDK, simulators, solution examples and development resources
- Range of hardware available on each platform, e.g. AWS Braket and Microsoft Azure Quantum
- Can be expensive to run jobs with many qubits and complex circuits



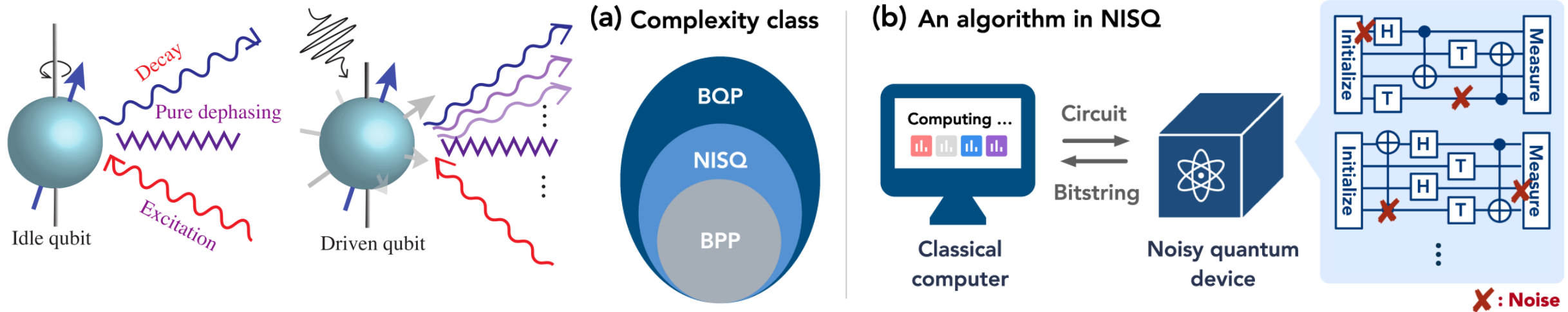
# NISQ - Noisy Intermediate-Scale Quantum Era

Physical qubits encounter noise due to interactions with external environment, hardware imperfections and cross-talk

Decoherence occurs when the physical qubit loses accuracy in the quantum state due to noise and decay

**NISQ (Noisy Intermediate-Scale Quantum) era** is the current state of quantum computing

Machines may have 100s/1000s of physical qubits but are limited by noise, decoherence and error



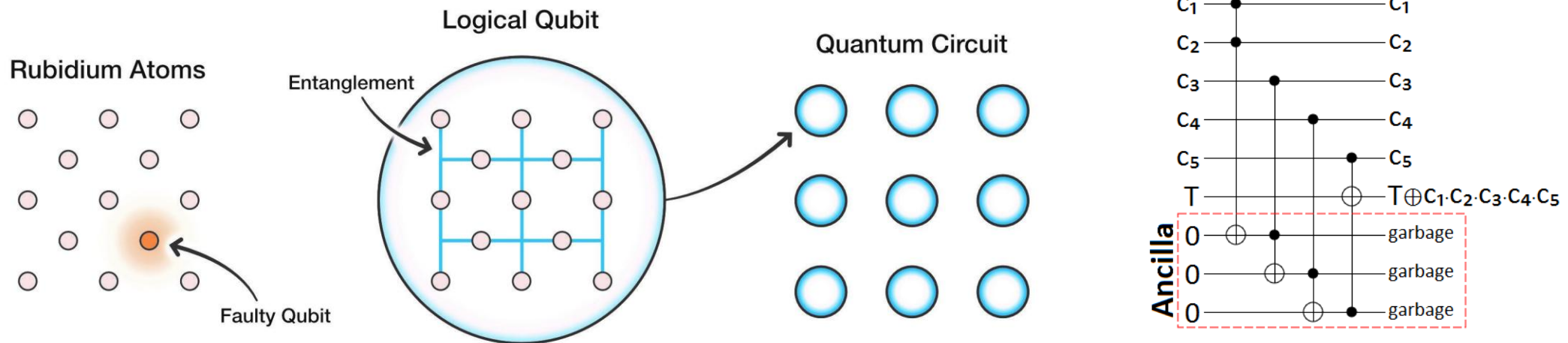
# Physical vs Logical and Ancilla Qubits

Physical qubits can be faulty due to noise and decoherence

Hence multiple physical qubits entangled together are used to create a “logical qubit” to represent information

Ancilla qubits enable control and operation of complex quantum gates, e.g Toffoli gates

Require “uncomputation” at the end to remove the effect of garbage information encoded there



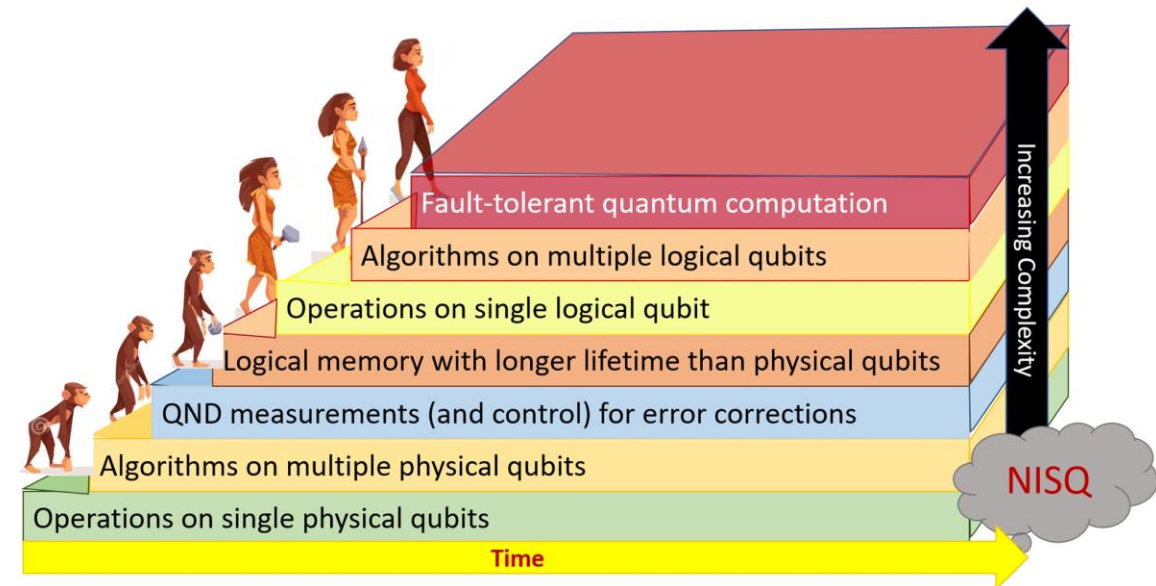
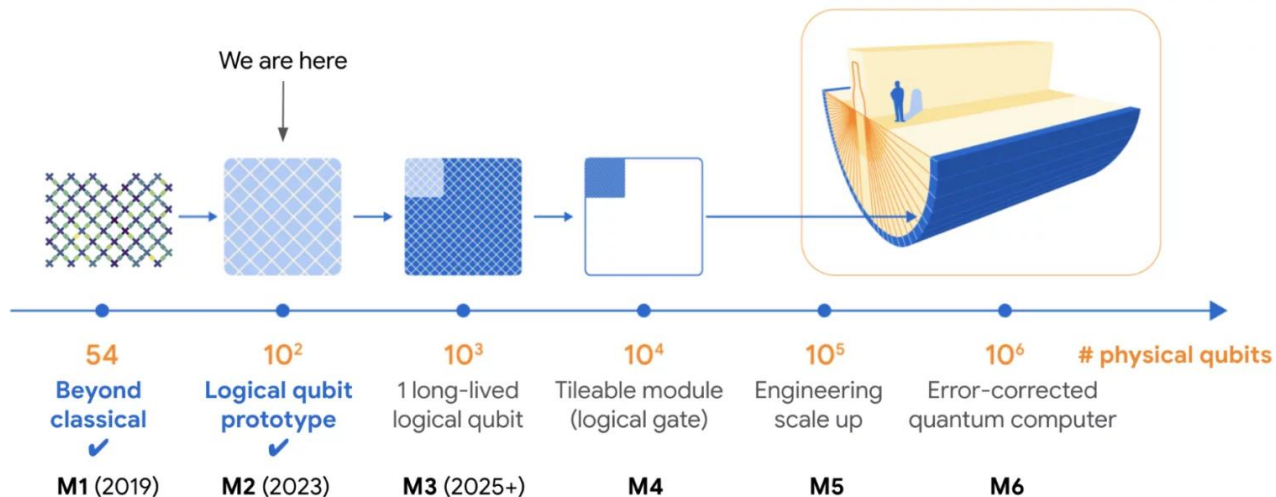
# Quantum Error Correction (QEC) and Fault Tolerant Computing

Goal: quantum computers which can correct errors in real time and retain accurate quantum states on physical qubits

Breakthroughs will enable scaling of algorithms to 1000s and 10,000s of qubits using multiple QPUs

Enables memory storage and access without conventional compute

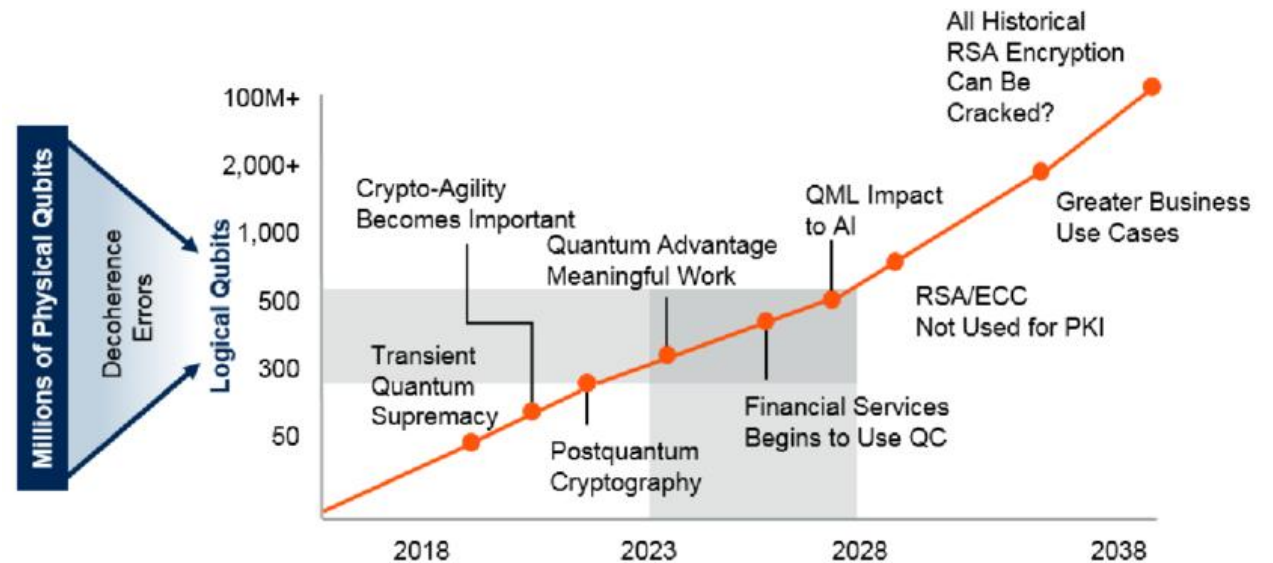
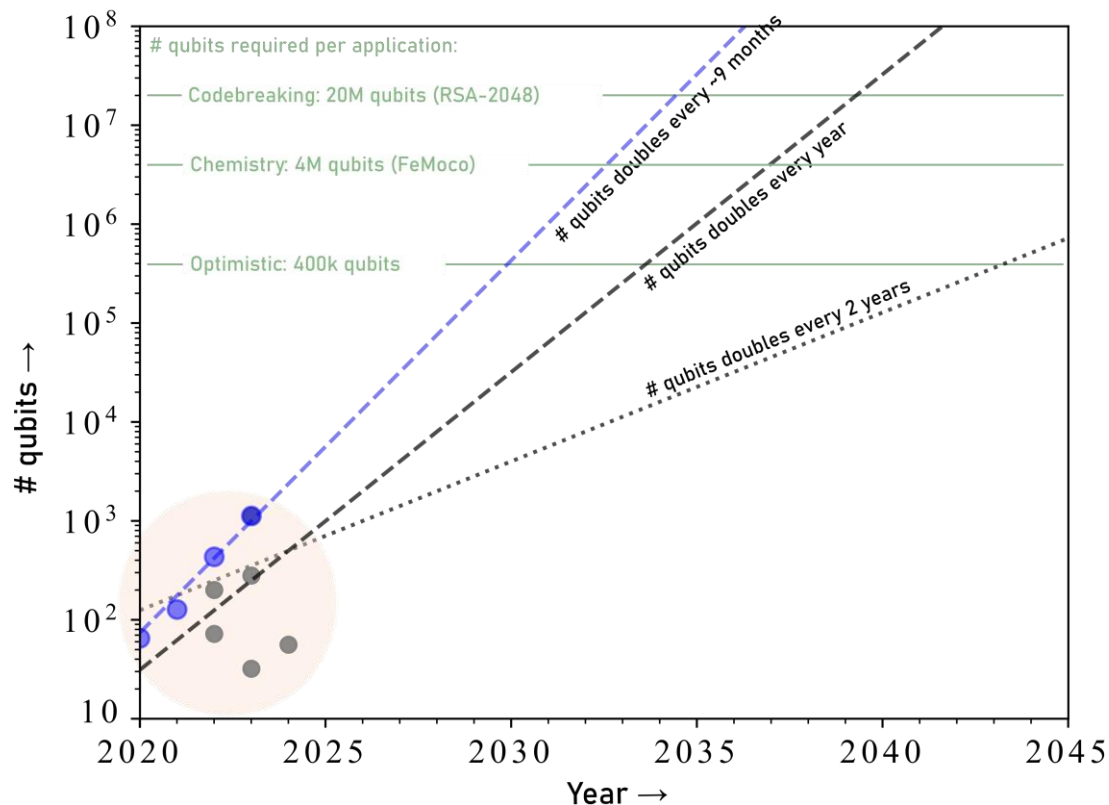
Active areas of research and investment, e.g. Google Willow and Microsoft Majorana



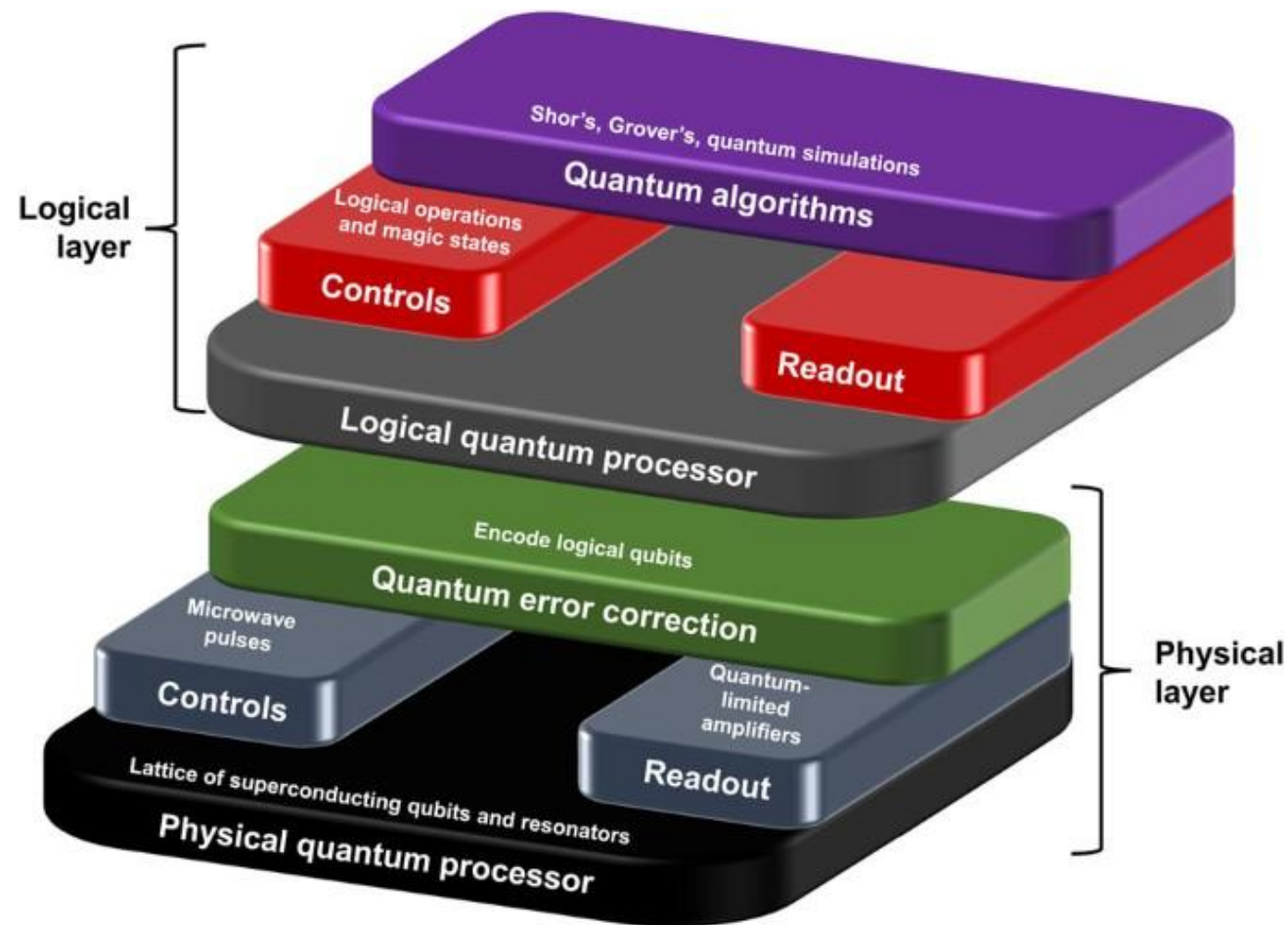
# Breakthroughs in Qubits and Applications

Quantum computing matures and makes fundamental breakthroughs in number of qubits and error correction

Qubit growth estimates, according to Moore's Law



# Quantum Computing Tech Stack





# The Quantum Computing Ecosystem

# The Quantum Computing Explosion



The quantum ecosystem expanded rapidly from 2020 onwards

<https://quantum2025.org>



User-level

Applications/Algorithms

Middleware

Hardware

## Developer & programming tools



## Cross-industry applications



## Superconducting circuits



## Photonics



## Neutral atoms



## Silicon, carbon, & helium



## Trapped ions



## Drug discovery



## Financial services



## Chemical & materials simulation



## Optimization & logistics



## Quantum hardware components



## Qubit control & error correction

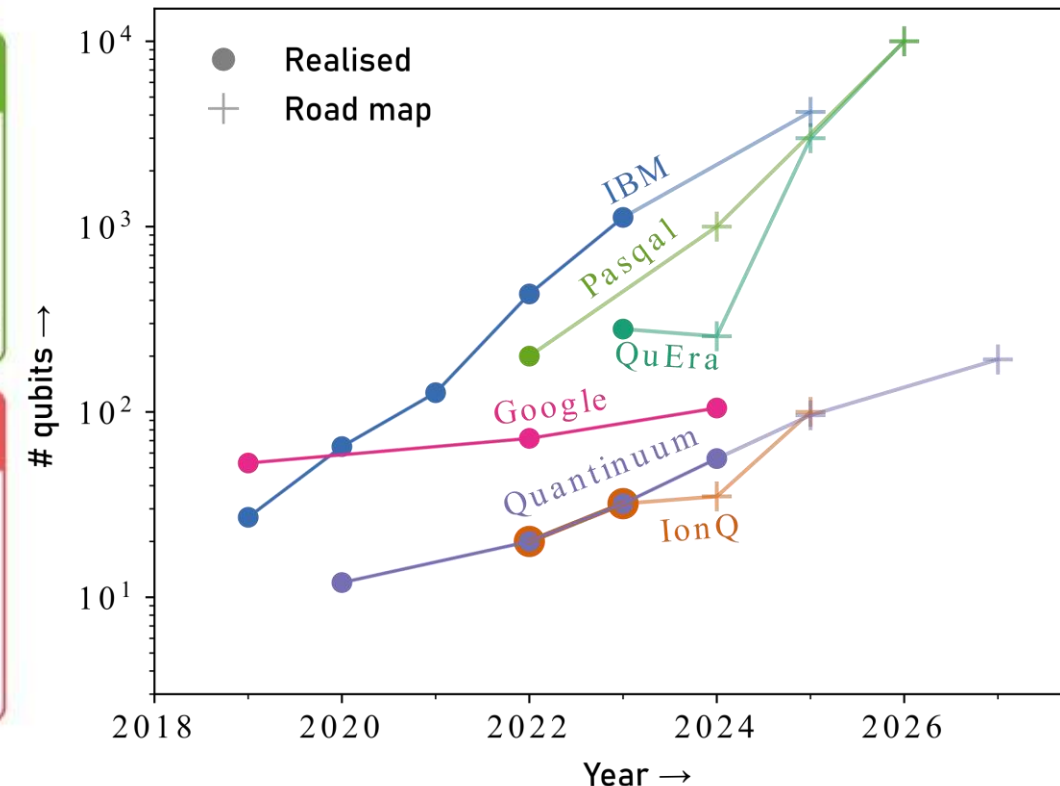
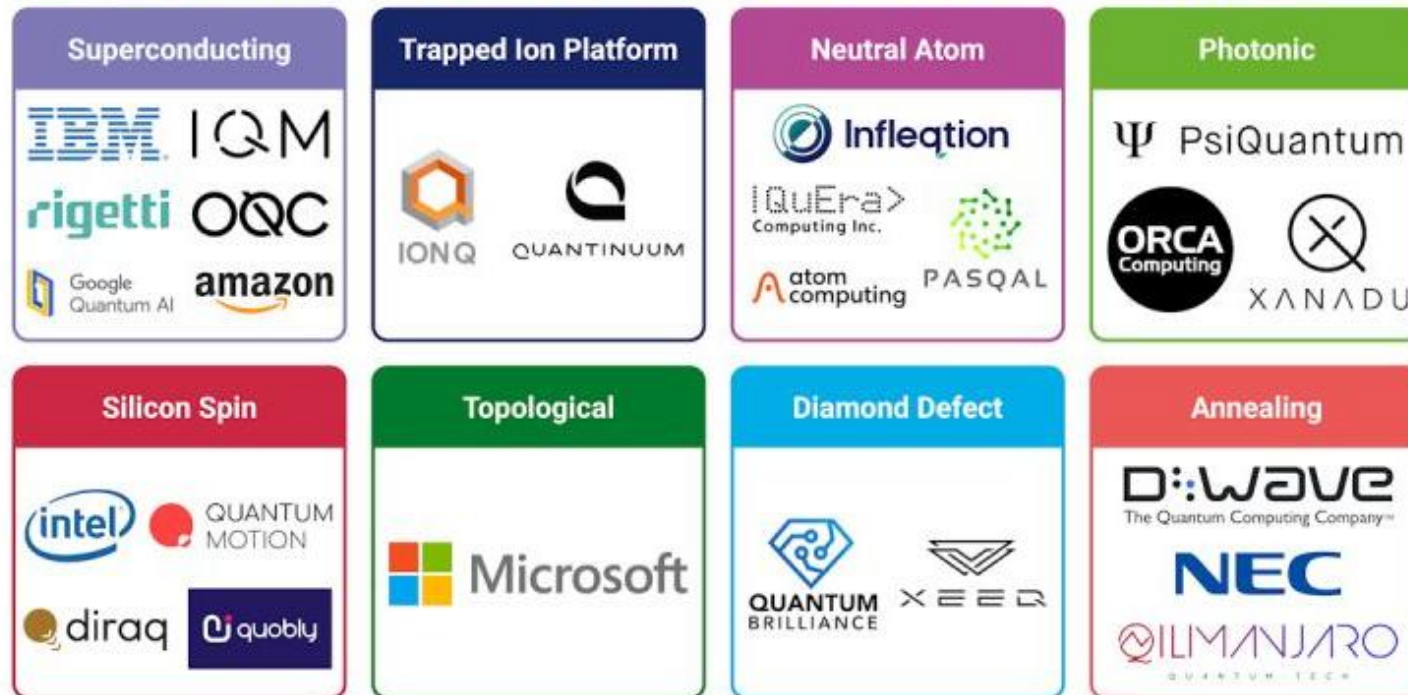


# Commercial Quantum Computers

~200 quantum computers in the world today

“Kiloqubit era” Number of qubits in the range of 1000s

IDTechEx Research **Eight Leading Approaches to Building a Commercial Quantum Computer**

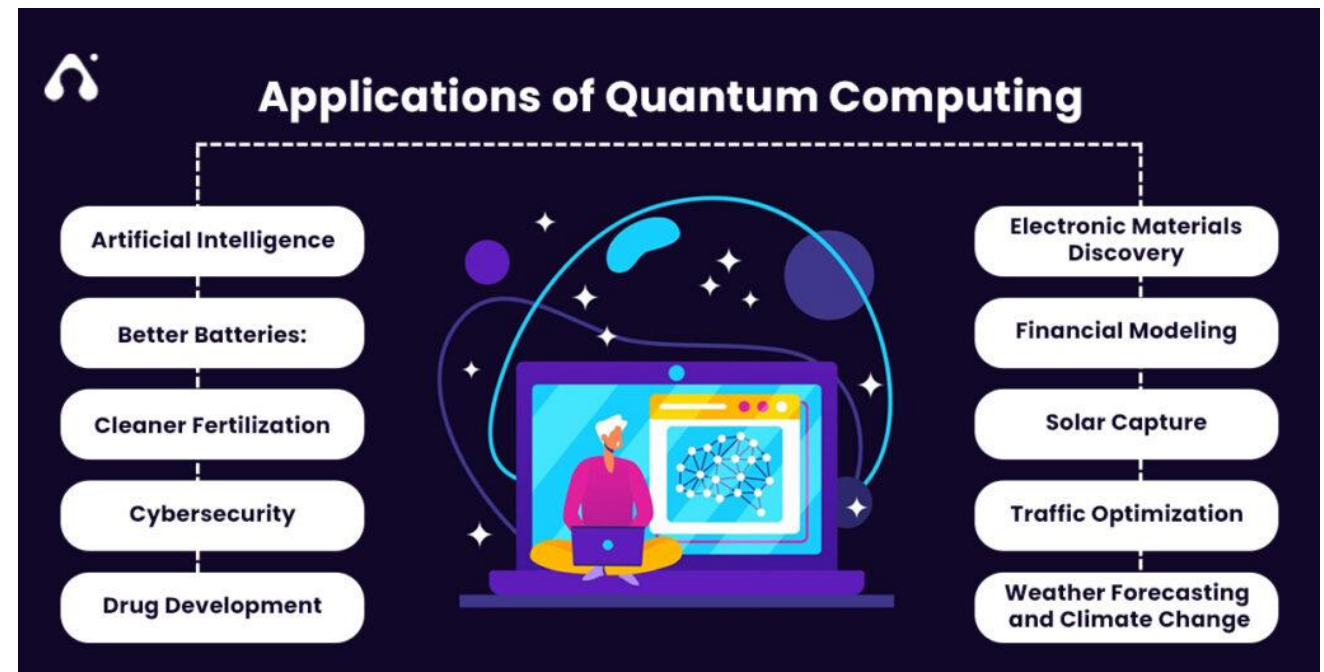
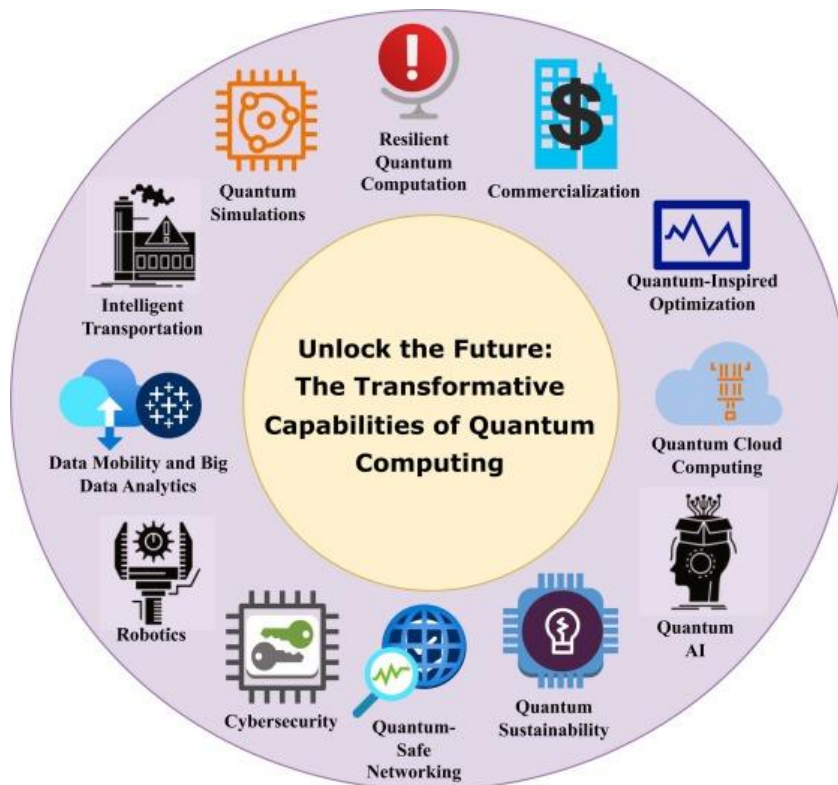


# Quantum System Applications

Applications of quantum computing (Native and inspired) are manifold and already in progress

Transportation and scheduling: airline scheduling, cargo loading, ports

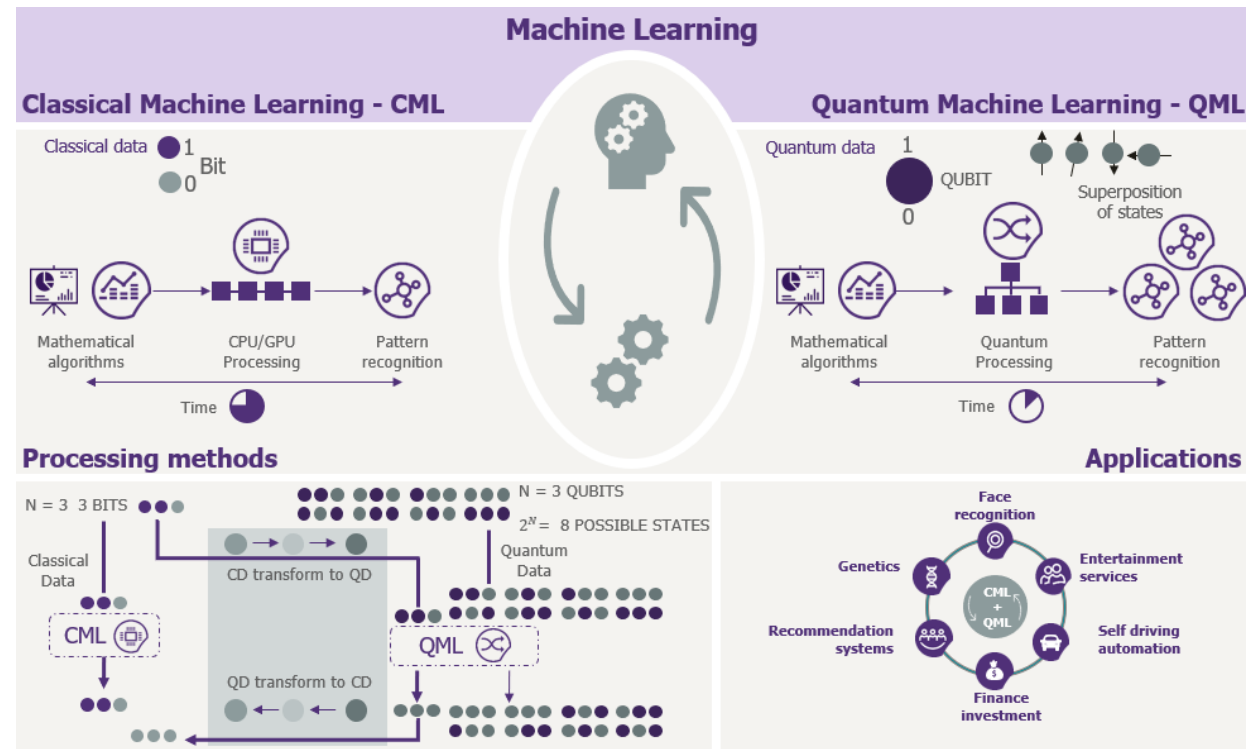
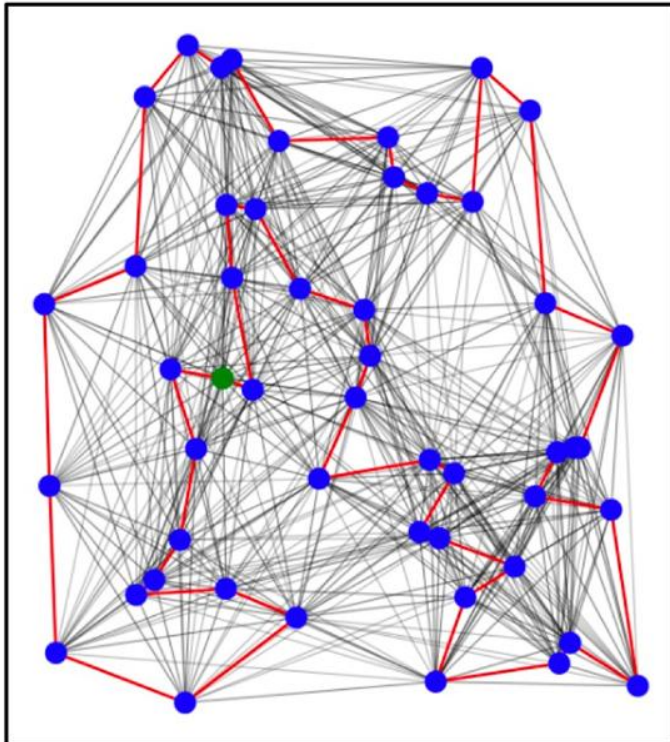
Cyber security: encryption, de-encryption, communication



# Quantum Optimization and Quantum Machine Learning

Quantum Optimization is highly suited to large scale solve systems modeling, scheduling and data processing problems

Quantum Machine Learning (QML) reduces model size, training accuracy and transfer learning to new data

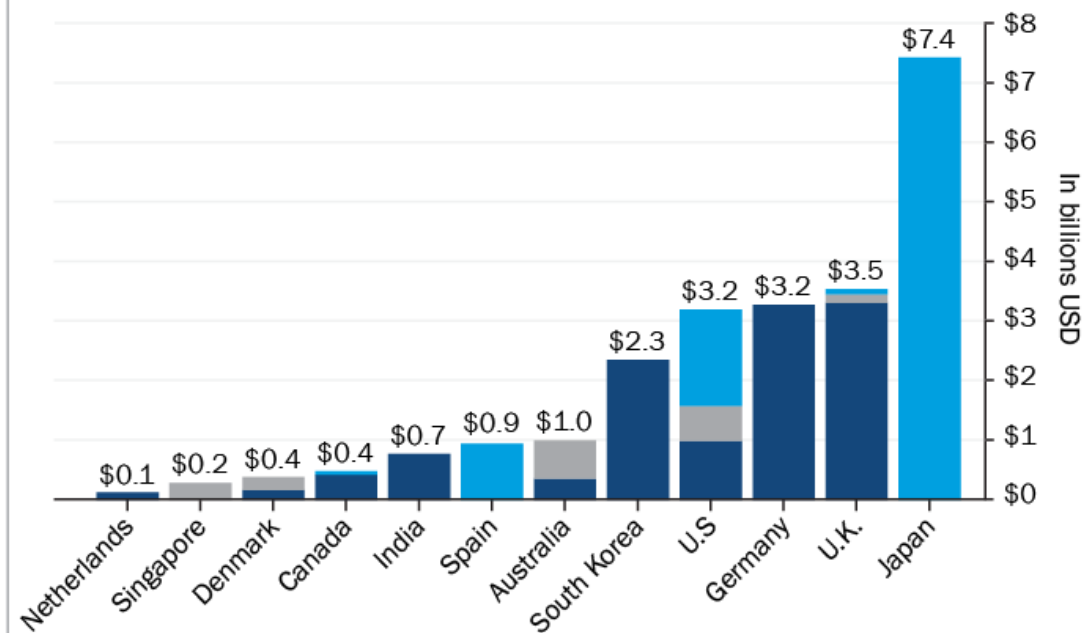


# Investments in Quantum Computing

Government Investments (> 10B), Private Investment (~\$2.2B)

## GOVERNMENT INVESTMENTS IN QUANTUM TECH

Year announced: ■ 2023 ■ 2024 ■ 2025 (Jan - Apr)



www.stansberryresearch.com

Source: McKinsey & Co., "The Year of Quantum," 2025

## Total private investment in Quantum Technology over the past 10 years

Total Quantum Investment by Stage; in \$ millions



Source: The Quantum Insider Intelligence Platform

# VC Deal Flow and Funding

Jump in 2025 – 111 deals, \$3.9B

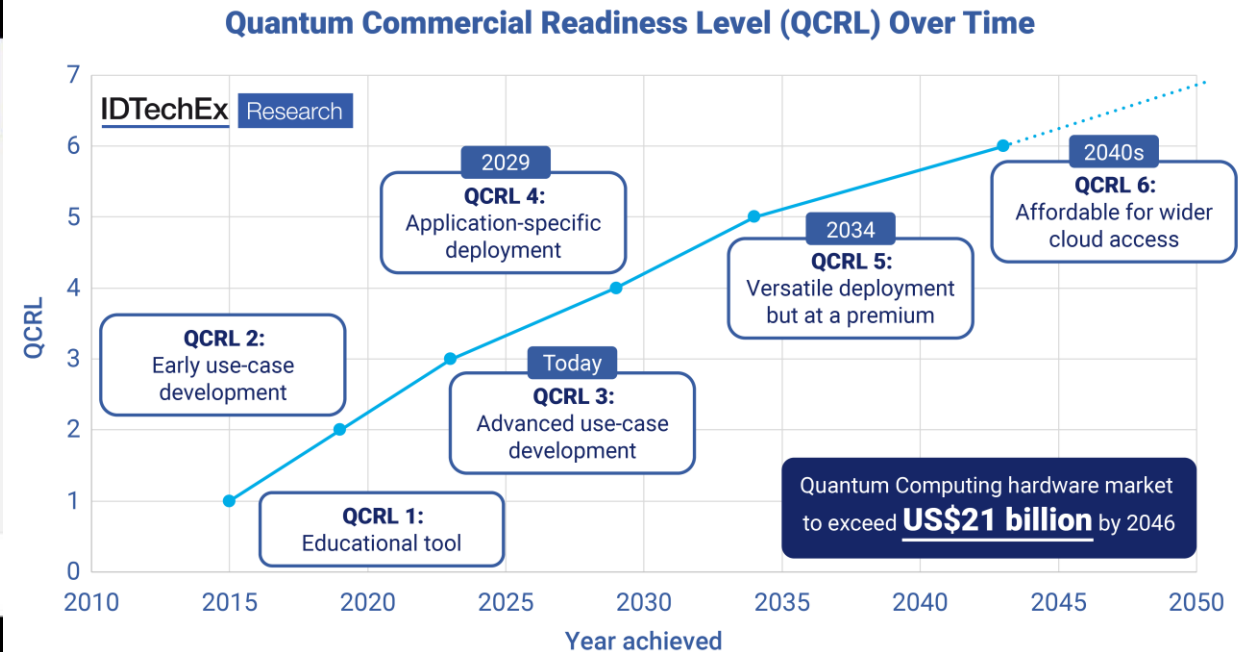
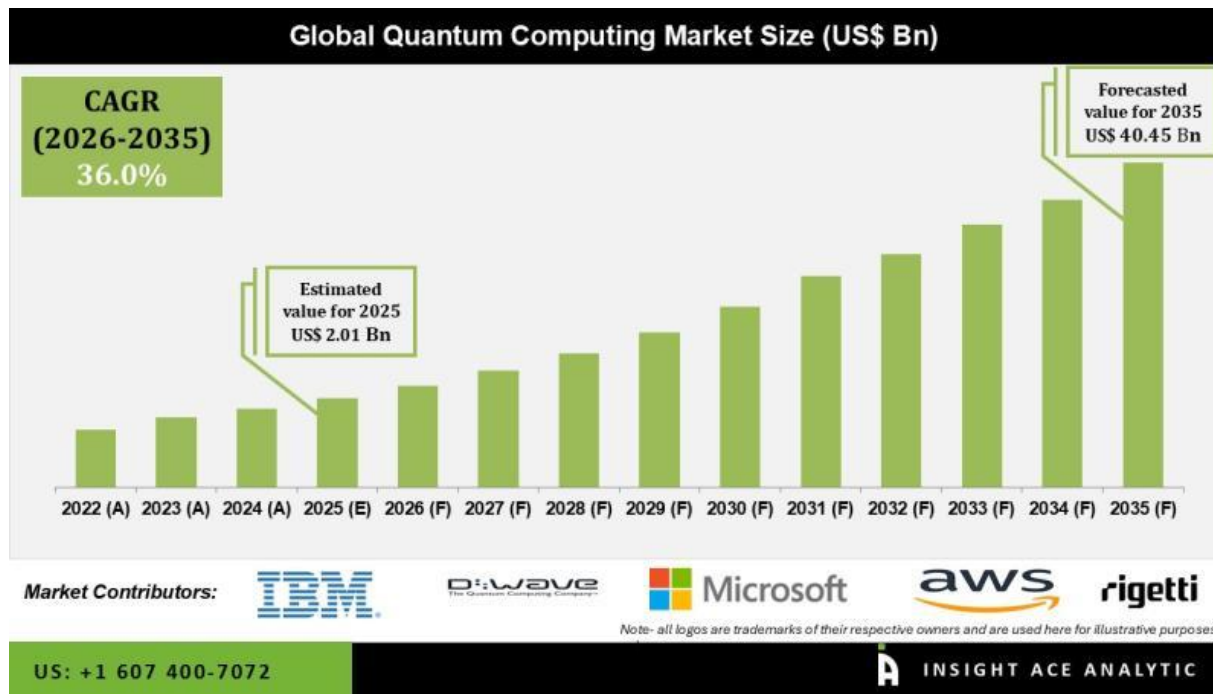
## Global quantum computing VC funding

Annually; 2020-2025 (as of Oct. 29)

|      | Total capital invested                                                                       | Deal count |
|------|----------------------------------------------------------------------------------------------|------------|
| 2025 |  \$3.9b    | 111        |
| 2024 |  \$1.3b     | 105        |
| 2023 |  \$1.3b    | 116        |
| 2022 |  \$1.1b   | 94         |
| 2021 |  \$1.3b   | 85         |
| 2020 |  \$475.5m | 54         |

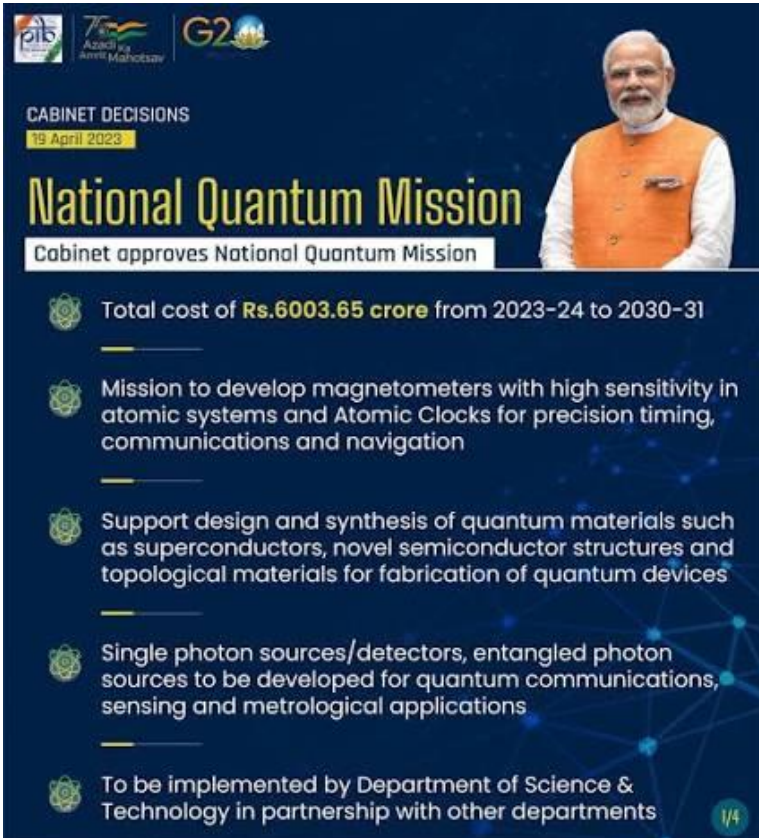
# Quantum Computing Market Evolution

Market evolving rapidly- current size is ~\$2 Billion, expected to grow 10-20x in coming decades



# NQM: Quantum Computing in India

India investing strategically via the National Quantum Mission started in 2023  
IITs and IISc serve as strategic hubs, State initiatives



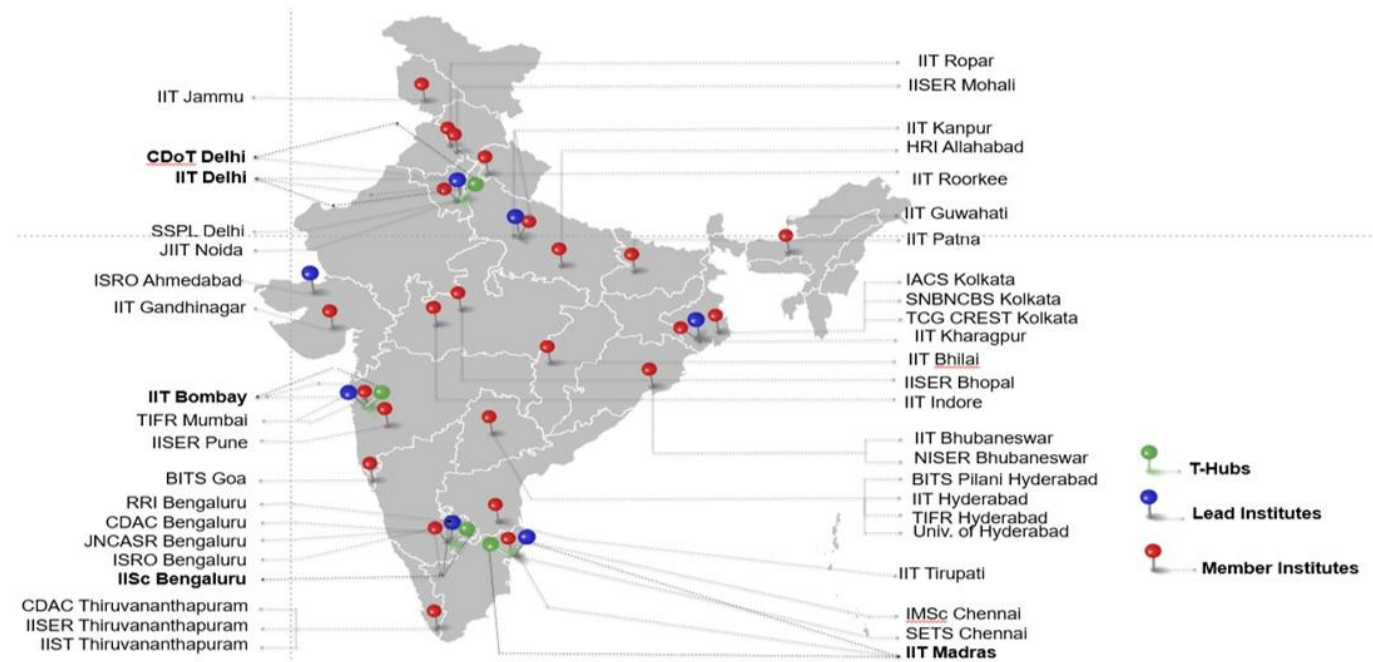
**CABINET DECISIONS**  
19 April 2023

**National Quantum Mission**  
Cabinet approves National Quantum Mission

- Total cost of **Rs.6003.65 crore** from 2023-24 to 2030-31
- Mission to develop magnetometers with high sensitivity in atomic systems and Atomic Clocks for precision timing, communications and navigation
- Support design and synthesis of quantum materials such as superconductors, novel semiconductor structures and topological materials for fabrication of quantum devices
- Single photon sources/detectors, entangled photon sources to be developed for quantum communications, sensing and metrological applications
- To be implemented by Department of Science & Technology in partnership with other departments

1/4

## Four Thematic Hubs Institutions supported under NQM



152 researchers from 43 Institutions (Lead & Member) across 17 states and 2 UTs involved in the 14 TGs

# References





DEMO

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THANK YOU!

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